



A New Alternative WENO Scheme Based on Exponential Polynomial Interpolation with an Improved Order of Accuracy

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Received: 1 November 2023 / Revised: 27 May 2024 / Accepted: 20 July 2024 /
Published online: 13 August 2024

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Abstract

In this study, we present a new alternative formulation of a conservative weighted essentially non-oscillatory (WENO) scheme that improves the performance of the known fifth-order alternative WENO (AWENO) schemes. In the formulation of the fifth-order AWENO scheme, the numerical flux can be written in two terms: a low-order flux and a high-order correction flux. The low-order numerical flux is constructed by a fifth-order WENO interpolator, and the high-order correction flux includes terms of the second and fourth derivatives, yielding the sixth-order truncation error. Noticing the difference in the convergence rates between these two approximations, this study first aims to fill the accuracy gap by enhancing the approximation order of the low-order numerical flux. To this end, the WENO interpolator for the low-order term is implemented using exponential polynomials with a shape parameter. Selecting a locally optimized shape parameter, the proposed WENO interpolator achieves an additional order of improvement, resulting in the overall sixth order of accuracy of the final reconstruction, under the same fifth-order AWENO framework. In addition, since a linear approximation to the high-order correction term may cause some oscillations in the vicinity of strong shocks, we present a new strategy for the limiting procedure to deal with the second derivative term in the high-order correction flux. Several numerical results for the well-known benchmark test problems confirm the reliability of our AWENO method.

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Keywords Alternative WENO · Hyperbolic conservation laws · Exponential polynomial · Shape parameter · Convergence order · Smoothness indicator

1 Introduction

In this paper, we study the finite difference alternative WENO scheme for solving hyperbolic conservation laws of the form

$$u_t + \nabla \cdot f(u) = 0, \quad (1)$$

where $f(u)$ is a flux function of the conserved quantity $u = u(x, t) : \Omega \times I \rightarrow \mathbb{R}$ with a bounded physical domain Ω and time interval I . The Eq. (1) is equipped with the initial function $u(x, 0) = u_0(x)$ and appropriate spatial boundary conditions. The class of WENO schemes has been a very successful methodology for solving hyperbolic problems that admit shocks, contact discontinuities, and complex smooth solutions. The first WENO scheme, an improved technique of ENO [21, 22], was introduced by Liu, Chan and Osher [41] based on the cell averages. Later, Jiang and Shu [28] proposed $(2r - 1)$ th-order WENO (termed as WENO-JS) schemes with smoothness indicators that measure the sum of the scaled L^2 -norm of all derivatives of local polynomial interpolants. As noticed by Henrick, Aslam, and Powers [26], a well-known disadvantage of WENO-JS is that there is a loss of accuracy at critical points. To overcome this shortcoming of WENO-JS, a mapping procedure (WENO-M) was developed to retain the optimal convergence order at critical points, but it incurs extra computational costs. Borges et al. [6] have designed a different type of nonlinear weights (WENO-Z) that achieves optimal convergence accuracy at critical points and reduces computational costs compared to the WENO-M method. Further, there have been lots of attempts to enhance the performance of these WENO techniques: fifth [13, 17, 20, 62] or higher-order WENO [3, 14, 18, 31, 32], hybrid compact WENO [43, 45], central WENO [9, 33, 35], and other WENO schemes using non-polynomial functions [5, 19, 23–25, 27, 42, 60, 61], and different weight designs [1, 2, 4, 7, 10, 16, 35–37, 42, 59, 63].

In recent years, alternative WENO (AWENO) schemes, originally introduced by Shu and Osher [48], have been successfully applied for solving hyperbolic conservation laws [8, 12, 29, 30, 38, 40, 47, 54, 55]. In the AWENO framework, the interpolation method is applied to the conservative variables, instead of interpolating the values of the flux function. It allows us to admit any monotone flux in the scalar equation and arbitrary approximate Riemann solvers (e.g., LF (Lax–Friedrichs) or HLLC [53] solvers) in the system case, while the classical schemes use smooth flux splitting functions. This is one of main advantages of the alternative WENO formulations. Although the AWENO formulation requires high computational costs, it was observed that the AWENO schemes generate higher resolution, less dissipation and dispersion errors than the known WENO schemes, in the problems containing shocks, rapid slopes and complicated fine structures.

This study aims to present a new formulation of the conservative AWENO scheme (hereafter termed AWENO-E) which improves the performance of the known fifth-order AWENO schemes. It is well-known [48] that the alternative form of the fifth-order numerical flux consists of two approximation terms

$$\hat{f}_{j+1/2} = \hat{f}_{j+1/2}^L + \hat{f}_{j+1/2}^H. \quad (2)$$

The low-order numerical flux $\hat{f}_{j+1/2}^L$ is constructed by a fifth-order WENO interpolation to the conservative variables. The high-order term $\hat{f}_{j+1/2}^H$ (referred to as a ‘correction term’)

can be obtained, via the Taylor expansion of the equation in (2), involving the second and fourth derivatives terms

$$\hat{f}_{j+1/2}^H = -\frac{1}{24}\Delta x^2 \frac{\partial^2 f}{\partial x^2} \Big|_{x_{j+1/2}} + \frac{7}{5760}\Delta x^4 \frac{\partial^4 f}{\partial x^4} \Big|_{x_{j+1/2}}.$$

The second and fourth derivatives (in the correction term) are usually approximated by a central finite difference method, yielding the overall sixth-order of accuracy, i.e., $\mathcal{O}(\Delta x^6)$ (e.g., see [48, 55]). It guarantees the fifth-order accuracy to the final reconstruction. But, one should notice that there is an accuracy gap between these two approximations regarding $\hat{f}_{j+1/2}^L$ and $\hat{f}_{j+1/2}^H$. The first term has fifth-order accuracy, whereas the latter one is sixth-order accurate. Motivated by this observation, the first objective of this study is to fill the gap of accuracy by enhancing the convergence order of $\hat{f}^L$ under the same fifth-order AWENO framework. To achieve this, we employ an alternative interpolation method using a set of exponential polynomials with a shape parameter, say λ . We especially present a method for selecting the parameter λ which is well-suited to local data feature. Therefore, the associated low-order flux has an improved order of accuracy, that is,

$$f_{j+1/2} = \hat{f}_{j+1/2}^L + \mathcal{O}(\Delta x^6).$$

On the other hand, it has been observed (e.g., see [8]) that when the higher-order derivatives are directly approximated by the linear central differences, this linear approximation may cause some oscillations near (strong) shocks. To circumvent this drawback, we add an extra limiting procedure controlling such artifacts, while maintaining the required high-order accuracy in smooth regions. To be more precise, we approximate the second derivative in the high-order correction term $\hat{f}_{j+1/2}^H$ by applying suitable nonlinear limiters which play a role of balancing regularities between smooth region and singular region. To verify the effectiveness of the proposed AWENO algorithm, the well-known benchmark test problems are simulated.

The rest of this paper is organized as follows. In Sect. 2, we review the classical (fifth-order) alternative WENO schemes for solving hyperbolic conservation laws. Section 3 is devoted for the construction a new AWENO scheme enhancing the known fifth-order WENO schemes. Experimental results on the benchmark test problems are provided in Sect. 4 and a conclusion is given in Sect. 5. Finally, we provide some required explicit formulas associated with the proposed AWENO scheme in Appendix.

2 WENO Schemes

2.1 Some Notations and Basic Notions

In this section, we present a general formulation associated with the finite difference alternative WENO framework for solving hyperbolic conservation laws in one-dimensional scalar case:

$$u_t + f(u)_x = 0, \quad (3)$$

where $u = u(x, t) : \Omega \times I \rightarrow \mathbb{R}$ with a bounded domain $\Omega = [a, b]$ and time interval I . The Eq. (3) is equipped with an initial condition $u(x, 0) = u_0(x)$. In this study, we suppose that the computation domain $[a, b]$ is partitioned uniformly, i.e., $x_n = a + n\Delta x$, $n = 0, \dots, N$, with $\Delta x = (b - a)/N$. The cell boundary $x_{j+1/2}$ can be written as $x_{j+1/2} = x_j + \frac{1}{2}\Delta x$. For a given function $f(u)$, f_n indicates the function value $f(u(x_n))$.

A semi-discrete conservative finite difference scheme for solving the equation in (3) is of the form

$$\frac{du_j(t)}{dt} = -\frac{1}{\Delta x}(\hat{f}_{j+1/2} - \hat{f}_{j-1/2}),$$

where $u_j(t)$ is a numerical approximation to the nodal value $u(x_j, t)$. Assuming that the flux f is a smooth function of u , the numerical flux $\hat{f}_{j+1/2}$ is designed to be a high-order approximation to $f(u)$ at the cell-interface $x_{j+1/2}$ such that

$$\frac{1}{\Delta x}(\hat{f}_{j+1/2} - \hat{f}_{j-1/2}) = f(u)_x|_{x=x_j} + \mathcal{O}(\Delta x^\gamma) \quad (4)$$

for a γ -th-order scheme. It is well-known in the literature [48] that, by using the Taylor expansion argument of the Eq. (4), a numerical flux with $(2r - 1)$ th-order accuracy can be obtained in the form:

$$\hat{f}_{j+1/2} = \hat{f}_{j+1/2}^L + \sum_{n=1}^{r-1} c_{2n} \Delta x^{2n} \left(\frac{\partial^{2n} f}{\partial x^{2n}} \right)_{j+1/2} + \mathcal{O}(\Delta x^{2r})$$

for some suitable constants c_{2n} . When $r = 3$, the coefficients c_n are given as $c_2 = -\frac{1}{24}$, $c_4 = \frac{7}{5760}$. The low-order approximation term $\hat{f}_{j+1/2}^L$ is obtained numerically by

$$\hat{f}_{j+1/2}^L = h(\hat{u}_{j+1/2}^-, \hat{u}_{j+1/2}^+)$$

with a bivariate monotone flux function h (e.g., Lax–Friedrichs flux), where $\hat{u}_{j+1/2}^\pm$ are the fifth-order (one-sided) WENO interpolations to u . In this alternative WENO framework, the interpolation method is applied to the conservative variables. It allows us to admit any monotone flux in the scalar equation and arbitrary approximate Riemann solvers (e.g., Lax–Friedrichs or HLLC solvers) in the system case. Moreover, to achieve the fifth-order approximation, the second and fourth derivatives may be approximated by

$$\Delta x^\alpha \partial_x^\alpha f|_{x_{j+1/2}} \approx \mathbb{D}_{S_6}^\alpha f(x_{j+1/2}), \quad \alpha = 2, 4, \quad (5)$$

where $\mathbb{D}_{S_6}^\alpha f(x_{j+1/2})$ is the α th-order central undivided difference of f on $S_6 = \{x_{j-2}, \dots, x_{j+3}\}$ satisfying

$$\mathbb{D}_{S_6}^\alpha f(x_{j+1/2}) := \Delta x^\alpha \partial_x^\alpha f|_{j+1/2} + \mathcal{O}(\Delta x^6),$$

provided that the function f is smooth enough. The specific form of $\mathbb{D}_{S_6}^\alpha f(x_{j+1/2})$ and its modification by a WENO limiter will be discussed later.

2.2 Fifth-Order AWENO Scheme for $\hat{u}^\pm$

A most important building block for a high-order alternative WENO scheme is the construction of the WENO interpolations $\hat{u}_{j+1/2}^-$ and $\hat{u}_{j+1/2}^+$. The framework of the fifth-order WENO interpolation procedure uses two five-point stencils S_-^5 and S_+^5 around the cell-interface $x_{j+1/2}$ for the construction of $\hat{u}_{j+1/2}^-$ and $\hat{u}_{j+1/2}^+$ respectively; see Fig. 1. In this study, for the sake of simplicity, we shall focus mainly on the construction of $\hat{u}^-$ on S_-^5 ; the other case can be done similarly (in fact, in a mirror symmetric way). Let

$$S_- := S_-^5(j) := \{x_{j-2}, \dots, x_{j+2}\} \quad (6)$$

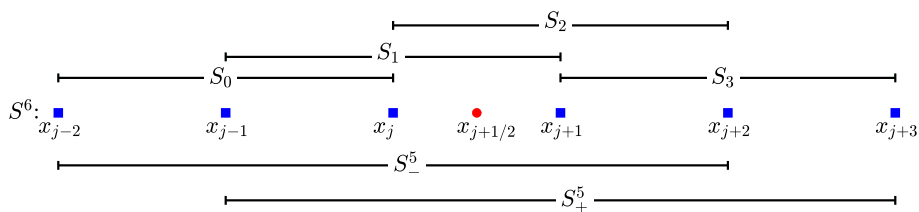


Fig. 1 Stencils $S_{\pm}^5$ and substencils $S_k, k = 0, \dots, 3$

be the left-biased five-point stencil around $x_{j+1/2}$. Here, for simplicity, we omit the index ‘5’ unless it causes confusion. With the purpose of constructing an essentially non-oscillatory numerical flux, the stencil S_- is divided into three candidate substencils:

$$S_k := S_{k,-}(j) := \{x_{j+k-2}, x_{j+k-1}, x_{j+k}\}, \quad k = 0, 1, 2. \quad (7)$$

On each substencil S_k , a local polynomial interpolation $\hat{u}^{[k]}$ of degree 2 is calculated and these local approximations are combined to obtain a high-order accuracy at the evaluation point $x_{j+1/2}$ (in smooth regions). Let $\hat{u}^-$ be the fourth-degree polynomial which interpolates the data given on the global stencil S_- . There are constants d_k with $k = 0, 1, 2$, called ideal (or linear) weights, such that the global interpolation $\hat{u}_{j+1/2}^-$ is written as

$$\hat{u}_{j+1/2}^- = \sum_{k=0}^2 d_k \hat{u}_{j+1/2}^{[k]}. \quad (8)$$

The specific values of d_k are known as $d_0 = \frac{1}{16}$, $d_1 = \frac{10}{16}$ and $d_2 = \frac{5}{16}$. Another important ingredient in the WENO interpolation procedure is the construction of smoothness indicators that measure the smoothness of the given data on the given stencil. According to these indicators, the ideal weights d_k are changed to the nonlinear weights ω_k . Then the final WENO reconstruction is defined by a convex combination of these local solutions:

$$\hat{u}_{j+1/2}^- = \sum_{k=0}^2 \omega_k \hat{u}_{j+1/2}^{[k]},$$

in a way of maintaining essentially non-oscillatory behavior around discontinuities while preserving optimal convergence order in smooth regions. The classical WENO-JS scheme [28] defines the nonlinear weights ω_k as

$$\omega_k := \frac{\alpha_k}{\sum_{\ell=0}^2 \alpha_{\ell}} \quad \text{and} \quad \alpha_k := \frac{d_k}{(\varepsilon + \beta_k)^2},$$

where a small positive value $\varepsilon > 0$ is employed to prevent the division by zero, and where the smoothness indicators β_k are defined by

$$\beta_k := \sum_{n=1}^2 \int_{x_{j-1/2}}^{x_{j+1/2}} \Delta x^{2n-1} \left(\frac{d^n u^{[k]}(x)}{dx^n} \right)^2 dx. \quad (9)$$

In a similar fashion, the WENO interpolation $\hat{u}_{j+1/2}^+$ is constructed on the right-biased five-point stencil

$$S_+ := S_+^5(j) := \{x_{j-1}, \dots, x_{j+3}\} \quad (10)$$

that is equivalent to shifting the stencil S_- to the right by one grid spacing. As in the case of (7), the substencils of S_+ are defined by

$$S_k := S_{k,+}(j) := \{x_{j+3-k}, x_{j+2-k}, x_{j+1-k}\}, \quad k = 0, 1, 2. \quad (11)$$

In addition, for later use, we introduce the notion of ‘the generalized undivided difference’ [20] which will be a useful tool for constructing new smoothness indicators as well as defining approximations to the α th-order derivatives $u^{(\alpha)}$ from a set of data values u_n on a given stencil.

• *Generalized undivided difference.* Let $x_{j+1/2}$ be a cell boundary and let $S := \{x_0, \dots, x_m\}$ be an $(m+1)$ -point stencil around $x_{j+1/2}$. Let v be a positive integer such that $v \leq m$. For the given values $u_n = u(x_n)$ with $x_n \in S$, we define the functional $\mathbb{D}_S^v u_{j+1/2}$ by

$$\mathbb{D}_S^v u_{j+1/2} := \sum_{x_n \in S} c_{v,n} u(x_n) \quad (12)$$

where the coefficients $c_{v,n}$ are obtained by solving the linear system

$$\sum_{x_n \in S} c_{v,n} \frac{(x_n - x_{j+1/2})^\alpha}{\alpha!} = \Delta x^\alpha \delta_{v,\alpha}, \quad \alpha = 0, \dots, m, \quad (13)$$

with $\delta_{v,\alpha}$ the Kronecker delta, i.e., $\delta_{v,\alpha}$ equals zero, if $v \neq \alpha$, and one otherwise. The coefficient vector $\mathbf{c}_v := (c_{v,n} : x_n \in S)^T$ of this linear system can be obtained by solving the matrix equation

$$\mathbf{V} \cdot \mathbf{c}_v = \mathbf{r}_v,$$

where $\mathbf{V}(\alpha, n) = (x_n - x_{j+1/2})^\alpha / \alpha!$ and $\mathbf{r}_v(\alpha) = \Delta x^\alpha \delta_{v,\alpha}$ for $\alpha, n = 0, \dots, m$. Since $\mathbf{V}$ is a Vandermonde matrix, this linear system is clearly unisolvant. Moreover, using the Taylor series argument, it can be shown that, as $\Delta x \rightarrow 0$,

$$\mathbb{D}_S^v u(x_{j+1/2}) = \Delta x^v u^{(v)}(x_{j+1/2}) + \mathcal{O}(\Delta x^{m+1}), \quad (14)$$

provided that the underlying function u is smooth enough. If $x_{j+1/2}$ is the central point of the stencil S , then $\mathbb{D}_S^v u(x_{j+1/2})$ becomes the v th-order central undivided finite difference of u on S . Also, if $v = m$, $\frac{1}{m!} \Delta x^{-m} \mathbb{D}_S^m u_{j+1/2}$ is the m th-order divided difference of u on S .

3 A New Formulation of the AWENO Scheme

Recall that the alternative form of the fifth-order numerical flux can be expressed as

$$\hat{f}_{j+1/2} = \hat{f}_{j+1/2}^L + \hat{f}_{j+1/2}^H,$$

where $\hat{f}_{j+1/2}^L$ is a low-order numerical flux constructed by a WENO interpolator and $\hat{f}_{j+1/2}^H$ is a high-order correction term. In this section, we present a new formulation of AWENO scheme which improves the performance of the known fifth-order AWENO schemes. To do this, we first propose a new WENO interpolation for f^L based on exponential polynomials. Next, we present a strategy of limiters for the error correction term $\hat{f}_{j+1/2}^H$.

3.1 WENO Interpolation for $\hat{f}^L$ Based on Exponential Polynomials

For the given point values u_i on a stencil, an interpolant $\hat{u}$ is computed from a certain approximation space. In this study, we construct an interpolation method based on a set

of exponential polynomials. In accordance with the framework of the fifth-order WENO scheme, we are particularly interested in the approximation spaces spanned by the following sets:

$$\begin{aligned}\mathbb{B}_5 &:= \{1, x, x^2, \sinh(\lambda x), \cosh(\lambda x)\}, \quad \lambda \in \mathbb{C}, \\ \mathbb{B}_3 &:= \{1, x, x^2\},\end{aligned}\tag{15}$$

which are associated to the interpolations on the global stencil S_- and its substencils S_k , $k = 0, 1, 2$, respectively. Clearly, when $\lambda \in i\mathbb{R}$, the exponential functions become trigonometric polynomials. In fact, for the given values on a stencil, finding a suitable value λ is an important issue to be investigated because the choice of λ has a strong impact to the accuracy of the final WENO reconstruction. For this reason, this study presents a method for selecting the shape parameter λ so that the associated approximation space would be more suited to the specific features of the given data. We will see that a carefully selected parameter allows the interpolation to achieve one additional order of accuracy, yielding the overall sixth-order of accuracy in the final approximation. This theoretical result will be confirmed by some numerical tests.

• *Global and local interpolation.* Let $S_- := \{x_{j-2}, \dots, x_{j+2}\}$ be the (left-biased) five-point stencil around the cell-interface $x_{j+1/2}$. Seeing that the space spanned by $\mathbb{B}_5$ is shift-invariant, the interpolation is practically defined using the translated basis functions, that is, $\mathbb{B}_5 = \{\phi_n(\cdot - x_{j+1/2}) : n = 0, \dots, 4\}$. Then, for the given values $u_i = u(x_i)$ with $x_i \in S_-$, we are seeking for an exponential polynomial from the space spanned by $\mathbb{B}_5$:

$$\mathcal{I}u = \sum_{n=0}^4 \alpha_n \phi_n(\cdot - x_{j+1/2}) \quad \text{s.t.} \quad \mathcal{I}u|_{S_-} = u|_{S_-}.$$

Then $\hat{u}^-$ at $x_{j+1/2}$ is defined by evaluating $\mathcal{I}u$ at $x_{j+1/2}$, i.e., $\hat{u}_{j+1/2}^- := \mathcal{I}u(x_{j+1/2})$. In particular, it is convenient to represent the interpolant $\mathcal{I}u$ as the following Lagrange-type:

$$\hat{u}_{j+1/2}^- = \mathcal{I}u(x_{j+1/2}) = \sum_{n=0}^4 \ell_n(x_{j+1/2}) u(x_{j-2+n}).\tag{16}$$

The vector of the Lagrange coefficients $\mathbf{L} := (\ell_n(x_{j+1/2}) : n = 0, \dots, 4)^T$ can be determined by solving the linear system

$$\mathbf{A} \cdot \mathbf{L} = \mathbf{R},\tag{17}$$

where the matrix $\mathbf{A}$ and the vector $\mathbf{R}$ have the components $\mathbf{A}(i, n) := \phi_i(x_{n-2} - x_{1/2})$ and $\mathbf{R}_n := \phi_n(0)$ for $i, n = 0, \dots, 4$, respectively. For better readability of this paper, the specific forms of the Lagrange coefficients $\ell_n(x_{j+1/2})$ are given in Appendix-(52).

Remark 3.1 Let $\mathbf{Q}$ be the vector of Lagrange coefficients of the classical fifth-order interpolation based on algebraic polynomials, that is, $\mathbf{Q} = (\frac{3}{128}, -\frac{5}{32}, \frac{45}{64}, \frac{15}{32}, -\frac{5}{128})$. Then, as presented in Appendix -(52), the vector $\mathbf{L}$ converges to $\mathbf{Q}$ in the sense of $\|\mathbf{L} - \mathbf{Q}\|_\infty = \mathcal{O}(\Delta x^2)$ as Δx tends to 0.

• *Local interpolation.* It can be inferred from (15) that the local interpolation $\hat{u}^{[k]} = \mathcal{I}_k u$ on each substencil S_k is defined by the second degree polynomial. This is the same as the case of the traditional fifth-order AWENO scheme and $\hat{u}^{[k]}$, $k = 0, 1, 2$, have the following

Lagrange type representations:

$$\begin{aligned}\hat{u}_{j+1/2}^{[0]} &= \frac{3}{8}u_{j-2} - \frac{5}{4}u_{j-1} + \frac{15}{8}u_j, \\ \hat{u}_{j+1/2}^{[1]} &= -\frac{1}{8}u_{j-1} + \frac{3}{4}u_j + \frac{3}{8}u_{j+1}, \quad \hat{u}_{j+1/2}^{[2]} = \frac{3}{8}u_j + \frac{3}{4}u_{j+1} - \frac{1}{8}u_{j+2}.\end{aligned}\quad (18)$$

• *Ideal weights based on exponential polynomials.* With the local approximations $\hat{u}_{j+1/2}^{[k]} = \mathcal{I}_k u(x_{j+1/2})$ at hand, as in the classical alternative WENO scheme in Sect. 2.2, we can find the so-called *linear weights* $\bar{d}_k \geq 0$, $k = 0, 1, 2$, with $\bar{d}_0 + \bar{d}_1 + \bar{d}_2 = 1$ such that its linear combination with the local solutions $\mathcal{I}_k u(x_{j+1/2})$ retrieves the global interpolation $\mathcal{I}u$ in (16) at the cell interface $x_{j+1/2}$:

$$\mathcal{I}u(x_{j+1/2}) = \sum_{k=0}^2 \bar{d}_k \mathcal{I}_k u(x_{j+1/2}).$$

In fact, since both the interpolation operators $\mathcal{I}$ and $\mathcal{I}_k$ reproduce the polynomials $\phi_n(x) = x^n$ for $n = 0, 1, 2$, the linear weights $\bar{d}_k$ are obtained by satisfying the following additional requirement

$$\mathcal{I}\phi_n(x_{j+1/2}) = \sum_{k=0}^2 \bar{d}_k \mathcal{I}_k \phi_n(x_{j+1/2}), \quad n = 3, 4, \quad (19)$$

where $\phi_3(x) = \sinh(\lambda x)$ and $\phi_4(x) = \cosh(\lambda x)$. Specifically, using the Lagrange functions in (16) and (18), the new ideal weights based on the exponential polynomials are defined by

$$\bar{d}_0 = \frac{8}{3}\ell_0(x_{j+1/2}), \quad \bar{d}_1 = 1 - \bar{d}_0 - \bar{d}_2, \quad \bar{d}_2 = -8\ell_4(x_{j+1/2}). \quad (20)$$

By applying the Taylor series argument to the Lagrange functions, it can be shown that, as $\Delta x \rightarrow 0$, the new weights $\bar{d}_k$ tend to the ideal weights $(\frac{1}{16}, \frac{10}{16}, \frac{5}{16})$ based on algebraic polynomials (see (8)). For more details, the readers are referred to the equations in Appendix- (53).

3.1.1 Analysis of Convergence Order

The aim of this section is to analyze the convergence order of the interpolation method based on exponential polynomials. We also suggest a method for finding a locally optimized shape parameter in the exponential functions. We will see that our choice of the shape parameter is related to the value $u^{(3)}(x_{j+1/2})$. So, our analysis considers the following two cases separately: $|u^{(3)}(x_{j+1/2})| \geq \kappa$ and $|u^{(3)}(x_{j+1/2})| < \kappa$ for some $\kappa > 0$.

Case I: $u^{(3)}(x_{j+1/2})$ is away from zero, i.e., $|u^{(3)}(x_{j+1/2})| \geq \kappa$ for some $\kappa > 0$.

Theorem 3.2 Assume that $u \in C^6(\Omega)$ with Ω an open set containing the stencil $S^6 = \{x_{j-2}, \dots, x_{j+3}\}$ around $x_{j+1/2}$. Let $\mathcal{I}u$ be the exponential polynomial in the space spanned by $\mathbb{B}_5$ such that $\mathcal{I}u|_{S_-} = u|_{S_-}$. Assume that $|u^{(3)}(x_{j+1/2})| \geq \kappa$ for some $\kappa > 0$ and λ in the exponential polynomial is chosen to satisfy the equation

$$\lambda^2 := \lambda_{j+1/2}^2 := \left(\frac{u^{(5)}}{u^{(3)}} \right)(x_{j+1/2}) + \mathcal{O}(\Delta x). \quad (21)$$

Then, there exists a suitable constant c_j depending on j and u such that

$$\mathcal{I}u(x_{j+1/2}) - u(x_{j+1/2}) = c_j \Delta x^6 + \mathcal{O}(\Delta x^7). \quad (22)$$

Proof For notational simplicity, denote $\bar{x} = x_{j+1/2}$. To facilitate our analysis, we first reorganize the exponential functions in $\mathbb{B}_5 = \{\phi_0, \dots, \phi_4\}$ as follows. Put $\bar{\phi}_n(x) := \phi_n(x)$ for $n = 0, 1, 2$ and

$$\bar{\phi}_3(x) := \frac{1}{\lambda^3} \left(\sinh(\lambda x) - \lambda x \right), \quad \bar{\phi}_4(x) := \frac{1}{\lambda^4} \left(\cosh(\lambda x) - 1 - \frac{(\lambda x)^2}{2} \right). \quad (23)$$

Then we define an auxiliary function ψ by

$$\psi := \sum_{n=0}^4 u^{(n)}(\bar{x}) \bar{\phi}_n(\cdot - \bar{x}). \quad (24)$$

(Note that the function ψ is employed only for the proof of this theorem, not for a practical computation.) From the definition of $\bar{\phi}_n$, it can be easily shown that $\bar{\phi}_\alpha^{(\beta)}(0) = \delta_{\alpha,\beta}$ for $\alpha, \beta = 0, \dots, 4$, where $\delta_{\alpha,\beta}$ indicates the Kronecker delta. Hence, it holds immediately from (24) that

$$\psi^{(\alpha)}(\bar{x}) = u^{(\alpha)}(\bar{x}), \quad \alpha = 0, \dots, 4. \quad (25)$$

Thus, using $u(\bar{x}) = \psi(\bar{x})$ and applying the exponential polynomial reproducing property in (17), we can induce the equation

$$u(\bar{x}) - \mathcal{I}u(\bar{x}) = \sum_{n=0}^4 \ell_n(\bar{x}) (\psi(x_{n+j-2}) - u(x_{n+j-2})). \quad (26)$$

To estimate the difference $\psi(x_{n+j-2}) - u(x_{n+j-2})$ on the right-hand side of the above equation, we use the Taylor expansion argument. For this, denote by T_g the 4th-degree Taylor polynomial of a smooth function g around $\bar{x}$. Then, by (25), it is apparent that $T_\psi = T_u$. Hence, seeing that $x_{n+j-2} - \bar{x} = \Delta x(n - 5/2)$, the remainder terms of these Taylor polynomials induce the expression

$$\psi(x_{n+j-2}) - u(x_{n+j-2}) = \Delta x^5 \frac{(n - 5/2)^5}{5!} (\psi^{(5)} - u^{(5)})(\bar{x}) + \mathcal{O}(\Delta x^6). \quad (27)$$

Here, it is easy to derive from the formula of ψ in (24) that $\psi^{(5)}(\bar{x}) = \lambda^2 u^{(3)}(\bar{x})$. Also, by the assumption in (21), we get $\lambda^2 u^{(3)}(\bar{x}) - u^{(5)}(\bar{x}) = \mathcal{O}(\Delta x)$. This together with (26) and (27) yields the identity

$$u(\bar{x}) - \mathcal{I}u(\bar{x}) = \Delta x^5 \left[\sum_{n=0}^4 \ell_n(\bar{x}) \frac{(n - 5/2)^5}{5!} (\lambda^2 u^{(3)} - u^{(5)})(\bar{x}) + \mathcal{O}(\Delta x) \right]. \quad (28)$$

As discussed in Remark 3.1, the Lagrange coefficient vector $\mathbf{L} = (\ell_n(\bar{x}) : n = 0, \dots, 4)$ tends to $\mathbf{Q}$ as $\Delta x \rightarrow 0$ with $\mathbf{Q}$ the Lagrange coefficient vector of the classical fifth order scheme. Thus, $\|\mathbf{L}\|_1 \leq \bar{c}$ for some constant $\bar{c} > 0$. Therefore, we obtain the required result in (22). $\square$

Remark 3.3 For a practical computation of the shape parameter λ satisfying (21), one may set λ as

$$\lambda^2 \Delta x^2 := \lambda_{j+1/2}^2 \Delta x^2 := \frac{\mathbb{D}_{S^6}^5 u_{j+1/2}}{\mathbb{D}_{S^6}^3 u_{j+1/2}},$$

provided that u is smooth enough on the stencil S^6 . Here the α th-order central undivided differences $\mathbb{D}_{S^6}^\alpha u_{j+1/2}$ for $\alpha = 3, 5$ are specifically defined by

$$\begin{aligned}\mathbb{D}_{S^6}^3 u_{j+1/2} &= (u_{j-2} - 13u_{j-1} + 34u_j - 34u_{j+1} + 13u_{j+2} - u_{j+3})/8, \\ \mathbb{D}_{S^6}^5 u_{j+1/2} &= -u_{j-2} + 5u_{j-1} - 10u_j + 10u_{j+1} - 5u_{j+2} + u_{j+3}.\end{aligned}\quad (29)$$

However, in case u is not smooth enough on S^6 , the value of λ may become large, yielding some undesirable artifacts. To circumvent this situation, one may assign an upper bound of λ , say $|\lambda| \leq |\lambda_0|$. Having experiences several alternatives, we found that a good choice is $|\lambda_0 \Delta x| = 0.25$.

Case II: $|u^{(3)}(x_{j+1/2})| \approx 0$, that is, $|u^{(3)}(x_{j+1/2})| < \kappa$ for some $\kappa > 0$.

Firstly, recall that $(q_n(\bar{x})) = (\frac{3}{128}, -\frac{5}{32}, \frac{45}{64}, \frac{15}{32}, -\frac{5}{128})$ are the Lagrange coefficients of the 4th-degree polynomial interpolation (see Remark 3.1). From the explicit formula of the Lagrange coefficient $\ell_n(\bar{x})$ in Appendix-(52), let us consider the following approximation

$$\ell_n(\bar{x}) \approx \ell_n^*(\bar{x}) := q_n(\bar{x}) + (\lambda \Delta x)^2 \ell_{n,2}(\bar{x})$$

with the coefficients $\ell_{n,2}(\bar{x})$ defined by $\{\ell_{n,2}(\bar{x})\} = \{-\frac{5}{1024}, \frac{1}{128}, \frac{3}{512}, -\frac{1}{64}, \frac{7}{1024}\}$, as given in Appendix-(52). Then, taking the Taylor expansion of u around $\bar{x} = x_{j+1/2}$, a direct calculation induces the expression

$$\sum_{n=0}^4 \ell_{n,2}(\bar{x}) u_{j-2+n} = -\frac{3}{256} \Delta x^3 u^{(3)}(\bar{x}) + \mathcal{O}(\Delta x^4).$$

Thus, when $u^{(3)}(\bar{x})$ is non-zero, by applying (21), the above arguments induce the equation

$$\sum_{n=0}^4 \ell_n^*(\bar{x}) u_{j-2+n} = \sum_{n=0}^4 q_n(\bar{x}) u_{j-2+n} - \frac{3}{256} \Delta x^5 u^{(5)}(\bar{x}), \quad (30)$$

where the terms on the right-hand side are indeed independent of $u^{(3)}(\bar{x})$. It means that the denominator $u^{(3)}(\bar{x})$ of λ^2 in (21) becomes a removable singularity. In this view point, for the case $|u^{(3)}(\bar{x})| < \kappa$ for some tolerance $\kappa > 0$, we modify the numerical flux $\hat{u}_{j+1/2}$ as

$$\hat{u}_{j+1/2}^- := \sum_{n=0}^4 \ell_n^*(\bar{x}) u_{j-2+n}. \quad (31)$$

Practically, for the given values u_i on S_- , the numerical flux $\hat{u}_{j+1/2}^-$ can be defined by

$$\hat{u}_{j+1/2}^- := \sum_{n=0}^4 q_n(\bar{x}) u_{j-2+n} - \frac{3}{256} \mathbb{D}_{S^6}^5 u_{j+1/2} \quad (32)$$

with the fifth-order central undivided difference $\mathbb{D}_{S^6}^5 u_{j+1/2}$ in (29). As a conclusion, we have the following result regardless of the issue of nullifying the denominator of λ .

Theorem 3.4 Assume that $u \in C^6(\Omega)$ with Ω an open set containing the stencil S_- . Let $\hat{u}_{j+1/2}^-$ be defined as in (31). Then, there exists a suitable constant c_j depending on j and u such that

$$\hat{u}_{j+1/2}^- - u(x_{j+1/2}) = c_j \Delta x^6 + \mathcal{O}(\Delta x^7).$$

Proof Taking the Taylor expansion of u_{j-2+n} around $\bar{x} = x_{j+1/2}$ and using the polynomial reproducing property of the coefficients q_n , $n = 0, \dots, 4$, up to degree ≤ 4 , it is immediate that

$$\sum_{n=0}^4 q_n(\bar{x}) u_{j-2+n} = u(\bar{x}) + \frac{3}{256} \Delta x^5 u^{(5)}(\bar{x}) + O(\Delta x^6).$$

This together with (30) leads to the required result. $\square$

In summary, the specific algorithm for choosing the shape parameter λ is given as follows.

Algorithm: Computation of the parameter λ

Input. The values u_n on S_- and the undivided differences $\mathbb{D}_{S_-}^3 u$, $\mathbb{D}_{S_-}^5 u$ at $x_{j+1/2}$.

Initialization. Put the thresholding value ξ as $\xi = \Delta x$.

0. If $|\Delta x^{-3} \mathbb{D}_{S_-}^3 u_{j+1/2}| \leq \xi$, then define the numerical flux $\hat{u}_{j+1/2}^-$ as given in (32)
 else put

$$T^2 := \frac{1}{\Delta x^2} \frac{\mathbb{D}_{S_-}^5 u_{j+1/2}}{\mathbb{D}_{S_-}^3 u_{j+1/2}}.$$

 end

1. If $|T^2| \geq 1$, then put $\lambda_{j+1/2} = \lambda_0$ for a fixed $\lambda_0 > 0$. Here we suggest $\lambda_0 \Delta x = 0.25$.

2. Otherwise (i.e., $|T^2| < 1$), if $T^2 > 0$, then $\lambda = |T|$. That is, put ϕ_3 and ϕ_4 as

$$\phi_3(x) = \sinh(|T|x) \quad \text{and} \quad \phi_4(x) = \cosh(|T|x).$$

 else if $T^2 < 0$, then $\lambda = i|T|$; put

$$\phi_3(x) = \sin(|T|x) \quad \text{and} \quad \phi_4(x) = \cos(|T|x).$$

 end

Here if T is zero, as discussed in Remark 3.1, the approximation space is considered as the classical algebraic polynomial space up to degree 4. It also yields the improved sixth-order accuracy.

3.2 Nonlinear Weights

In general, the smoothness indicators of a WENO scheme are designed to have certain decaying rates in smooth regions. It ends up that the local stencils containing a singularity are penalized such that their contributions to the final reconstruction are essentially zero. Following the strategy in [20], our smoothness indicators are constructed by applying the generalized undivided differences directly to the data u_i on the given stencil. First, the local smoothness indicators on S_k , $k = 0, 1, 2$, are define by

$$\beta_k := \theta \left| \mathbb{D}_{S_k} u_{j+1/2} \right| + \left| \mathbb{D}_{S_k}^2 u_{j+1/2} \right|, \quad \theta \in (0, 1], \quad (33)$$

where $\mathbb{D}_{S_k}^\alpha$ for $\alpha = 1, 2$ are expressed explicitly as

$$\begin{aligned} \mathbb{D}_{S_k} u &:= (1-k)u_{j-2+k} + (2k-3)u_{j-1+k} + (2-k)u_{j+k}, \\ \mathbb{D}_{S_k}^2 u &:= u_{j-2+k} - 2u_{j-1+k} + u_{j+k}. \end{aligned} \quad (34)$$

The value θ is a compromising balanced tradeoff between $\mathbb{D}_{S_k} u$ and $\mathbb{D}_{S_k}^2 u$, because these terms have different vanishing moments in a smooth area. In our numerical experiments, we set $\theta = 0.4$ for all test problems. Moreover, the regularity of the data on the global stencil S_- is defined by using the fourth-order undivided difference:

$$\tau_5 := |\mathbb{D}_{S_-}^4 u_{j+1/2}| = |u_{j-2} - 4u_{j-1} + 6u_j - 4u_{j+1} + u_{j+2}|.$$

Finally, according to the framework of the WENO-Z scheme, our nonlinear weights $\omega_k = \omega_k^-$ on S_- take the form

$$\omega_k = \frac{\alpha_k}{\sum_{\ell=0}^2 \alpha_\ell}, \quad \alpha_k = d_k \left(1 + \frac{\tau_5^p}{\beta_k^p + \epsilon} \right), \quad p \geq 1 \quad (35)$$

for $k = 0, 1, 2$. The number $\epsilon := \epsilon(\Delta x) > 0$ is employed to avoid nullifying denominator. It plays an important role to achieve the maximal accuracy of the final WENO reconstruction especially at the critical points. Also, in our numerical simulations, we set $p = 2$. The process to obtain nonlinear weights $\omega_k = \omega_k^+$ on the stencil $S_+ = S_+^5$ can be done in a way of mirror symmetric to that of (35) with respect to the location $x_{j+1/2}$.

Remark 3.5 As discussed in [55], the analysis about the convergence order of a AWENO scheme (especially, near the critical points) is the same as the case of the classical WENO schemes. Due to [31], a sufficient condition for a numerical flux to get the sixth convergence order is given as

$$\omega_k^\pm - \bar{d}_k = O(\Delta x^4), \quad (36)$$

where superscript ‘ $\pm$ ’ on the weight ω_k corresponds to $f_{j\pm 1/2}^k$ respectively. Thus, to compute $\omega_k^\pm - \bar{d}_k$ with the proposed weights $\omega_k^\pm$, let us assume that ϵ in the definition of α_k is chosen as $\epsilon = \Delta x^\gamma$ with $0 < \gamma \leq 4$. Then, since $\tau = O(\Delta x^4)$, the condition (36) can be derived immediately, regardless of the issue of critical points, provided that u is smooth enough around the global stencil S^6 .

Below, we summarize the algorithm for the AWENO interpolation based on exponential polynomials.

Algorithm: Computation of $u_{j+1/2}^\pm$ by AWENO-E($\{u_{j-2}, \dots, u_{j+3}\}, \lambda_0 \Delta x$),

1. Evaluate the undivided differences of u on the associated stencils: $\mathbb{D}_{S_-}^3 u$, $\mathbb{D}_{S_+}^3 u$, $\mathbb{D}_{S_-}^4 u$, $\mathbb{D}_{S_+}^4 u$ and $\mathbb{D}_{S^6}^5 u$ at $x = x_{j+1/2}$.
2. Determine the shape parameter λ ; see the Algorithm I.
3. Evaluate the ideal weights $\bar{d}_k$ in (20) (see also (53)) and nonlinear weights $\omega_k^\pm$ with $\beta_k^\pm$ as in (35) and (33) for global stencils $S_\pm$.
4. Using the formula in (18), obtain $\mathcal{I}_k^\pm u(x_{j+1/2})$ on the substencils $S_{k,\pm}$ in (7) and (11) for $k = 0, 1, 2$.
5. Construct exponential AWENO interpolation at the cell interface $x_{j+1/2}$:

$$u_{j+1/2}^\pm = \omega_0^\pm \mathcal{I}_0^\pm u(x_{j+1/2}) + \omega_1^\pm \mathcal{I}_1^\pm u(x_{j+1/2}) + \omega_2^\pm \mathcal{I}_2^\pm u(x_{j+1/2}).$$

3.3 Limiters for the Error Correction Term $\hat{f}^H$

If the higher-order derivatives are approximated directly by the central differences, the algorithm becomes unstable, resulting in unphysical negative pressures or some oscillations in the vicinity of strong shocks. (e.g., see Fig. 14 in Appendix). To circumvent this trouble, we add an extra limiting procedure controlling such artifacts, while maintaining the required high-order accuracy in smooth regions. To be more precise, we approximate the second derivative in the high-order correction term $\hat{f}_{j+1/2}^H$ by applying suitable nonlinear limiters which play a role of balancing regularities between smooth region and singular region. To do this, recall that $S_- = S_-^5$ and $S_+ = S_+^5$ (see (6) and (10)). Then, if f is smooth on these stencils, we can write

$$\Delta x^2 f''|_{x_{j+\frac{1}{2}}} := \Delta x^2 \frac{\partial^2 f}{\partial x^2}|_{x_{j+\frac{1}{2}}} \approx \mathbb{D}_{S_0}^2 f_{j+\frac{1}{2}} = \frac{1}{2}(\mathbb{D}_{S_-}^2 f_{j+\frac{1}{2}} + \mathbb{D}_{S_+}^2 f_{j+\frac{1}{2}}), \quad (37)$$

where $\mathbb{D}_{S_-}^2 f_{j+\frac{1}{2}}$ and $\mathbb{D}_{S_+}^2 f_{j+\frac{1}{2}}$ are constructed as in (12)–(14). Specifically, they are defined by

$$\begin{aligned} \mathbb{D}_{S_-}^2 \hat{f}_{j+\frac{1}{2}}^- &= -\frac{5}{24} f_{j-2} + \frac{4}{3} f_{j-1} - \frac{7}{4} f_j + \frac{1}{3} f_{j+1} + \frac{7}{24} f_{j+2}, \\ \mathbb{D}_{S_+}^2 \hat{f}_{j+\frac{1}{2}}^+ &= \frac{7}{24} f_{j-1} + \frac{1}{3} f_j - \frac{7}{4} f_{j+1} + \frac{4}{3} f_{j+2} - \frac{5}{24} f_{j+3}. \end{aligned} \quad (38)$$

However, since the stencils for the above central differences may include singularities, our idea for dealing with the approximation to the second derivative is to decompose each central finite difference in (38) as a linear combination of the second order undivided differences on the substencils S_k . To be more precise, letting

$$\mathbb{D}_{S_k}^2 f_{j+\frac{1}{2}} := f_{j-2+k} - 2f_{j-1+k} + f_{j+k},$$

the equations in (38) can be represented as

$$\begin{aligned} \mathbb{D}_{S_-}^2 \hat{f}_{j+\frac{1}{2}}^- &= \gamma_0 \mathbb{D}_{S_0}^2 f_{j+\frac{1}{2}} + \gamma_1 \mathbb{D}_{S_1}^2 f_{j+\frac{1}{2}} + \gamma_2 \mathbb{D}_{S_2}^2 f_{j+\frac{1}{2}}, \\ \mathbb{D}_{S_+}^2 \hat{f}_{j+\frac{1}{2}}^+ &= \gamma_2 \mathbb{D}_{S_1}^2 f_{j+\frac{1}{2}} + \gamma_1 \mathbb{D}_{S_2}^2 f_{j+\frac{1}{2}} + \gamma_0 \mathbb{D}_{S_3}^2 f_{j+\frac{1}{2}} \end{aligned} \quad (39)$$

with the coefficients γ_k defined by

$$\gamma_0 := -\frac{5}{24}, \quad \gamma_1 := \frac{22}{24}, \quad \text{and} \quad \gamma_2 := \frac{7}{24}. \quad (40)$$

Clearly, $\sum_{k=0}^2 \gamma_k = 1$. Then, our next task is to apply a suitable limiter to each coefficient γ_k for $k = 0, 1, 2$. However, since the linear weight $\gamma_0 = -\frac{5}{24}$ is negative, the limiting procedure may not be applied directly to obtain a stable scheme. To treat the negative weight, we suggest the following limiting procedure. First, as in (33), let $\beta_{f,k}$ be the smoothness indicator to the data values on the substencil S_k :

$$\beta_{f,k} = \theta |\mathbb{D}_{S_k} f_{j+1/2}| + |\mathbb{D}_{S_k}^2 f_{j+1/2}|, \quad \theta \in (0, 1]. \quad (41)$$

Noting the negative coefficient $\gamma_0 = -\frac{5}{24}$ is associated to the substencil S_0 , we present a tool which determines whether a singularity exists in the substencil S_0 or not. Specifically, put

$$\zeta := \begin{cases} -1, & \text{if } \beta_{f,0} < \frac{1}{2} \max(\beta_{f,1}, \beta_{f,2}), \\ 1, & \text{otherwise.} \end{cases} \quad (42)$$

If $\zeta = -1$, we infer that a singularity exists in S_- but not in S_0 , that is, S_0 is a smooth region. Otherwise, S_0 contains a singular point or the (global) stencil S_- is a smooth region. Then, letting

$$\sigma = \zeta \cdot \tilde{\alpha}_0 + \tilde{\alpha}_1 + \tilde{\alpha}_2 \quad (43)$$

with

$$\tilde{\alpha}_k = \gamma_k \left[1 + \left(\frac{|\mathbb{D}_{S_-}^4 f_{j+1/2}|^2}{\beta_k^2 + \epsilon} \right) \right], \quad k = 0, 1, 2, \quad (44)$$

and γ_k in (40), the limiters $\tilde{\omega}_k$ on the substencils S_k are defined by

$$\tilde{\omega}_0 = \frac{\zeta \cdot \tilde{\alpha}_0}{\sigma} \quad \text{and} \quad \tilde{\omega}_k = \frac{\tilde{\alpha}_k}{\sigma} \quad (k = 1, 2).$$

Consequently, using the limiters $\tilde{\omega}_k$, our nonlinear counterparts of $\mathbb{D}_{S_-}^2 \hat{f}_{j+\frac{1}{2}}^-$ and $\mathbb{D}_{S_+}^2 \hat{f}_{j+\frac{1}{2}}^+$ are defined as follows:

$$\begin{aligned} \Delta x^2 \widehat{f}_{j+\frac{1}{2}}''^- &= \tilde{\omega}_0 \mathbb{D}_{S_0}^2 f_{j+\frac{1}{2}} + \tilde{\omega}_1 \mathbb{D}_{S_1}^2 f_{j+\frac{1}{2}} + \tilde{\omega}_2 \mathbb{D}_{S_2}^2 f_{j+\frac{1}{2}}, \\ \Delta x^2 \widehat{f}_{j+\frac{1}{2}}''^+ &= \tilde{\omega}_2 \mathbb{D}_{S_1}^2 f_{j+\frac{1}{2}} + \tilde{\omega}_1 \mathbb{D}_{S_2}^2 f_{j+\frac{1}{2}} + \tilde{\omega}_0 \mathbb{D}_{S_3}^2 f_{j+\frac{1}{2}}. \end{aligned} \quad (45)$$

The above nonlinear approximations enhance the performance of the final WENO reconstruction. In fact, one may use a hybrid method (e.g., [39]) to reduce the computation time of implementing (45). In what follows, we discuss the convergence property of the nonlinear approximation (45).

Theorem 3.6 Assume that $f \in C^6(\Omega)$ with Ω an open set containing the stencils S_- and S_+ around $x_{j+1/2}$. Assume further that the parameter ϵ in $\tilde{\alpha}_k$ is chosen as $\epsilon = \Delta x^r$ with $0 \leq r \leq 4$. Then, we have

$$\Delta x^2 \frac{\partial^2 f}{\partial x^2} \Big|_{x_{j+\frac{1}{2}}} = \frac{\Delta x^2}{2} \left(\widehat{f}_{j+\frac{1}{2}}''^- + \widehat{f}_{j+\frac{1}{2}}''^+ \right) + \mathcal{O}(\Delta x^6).$$

Proof Recall that the second-order central difference of f on the stencil S^6 yields the sixth-order of a truncation error, that is

$$\Delta x^2 \frac{\partial^2 f}{\partial x^2} \Big|_{x_{j+\frac{1}{2}}} - \mathbb{D}_{S^6}^2 f_{j+\frac{1}{2}} = \mathcal{O}(\Delta x^6).$$

Then, due to the identity $\mathbb{D}_{S^6}^2 f_{j+\frac{1}{2}} = \frac{1}{2}(\mathbb{D}_{S_-}^2 f_{j+\frac{1}{2}} + \mathbb{D}_{S_+}^2 f_{j+\frac{1}{2}})$, it suffices to show that

$$\Delta x^2 \widehat{f}_{j+\frac{1}{2}}''^\pm - \mathbb{D}_{S_\pm}^2 f_{j+\frac{1}{2}} = \mathcal{O}(\Delta x^6). \quad (46)$$

To do this, we see that each β_k converges to the same limit, as $\Delta x \rightarrow 0$, because f is smooth on Ω . So, it can be easily inferred that $\zeta = 1$ (see (42)) for a certain small Δx . Moreover, by construction, $\beta_{f,k} = \mathcal{O}(\Delta x)$ and $\tau = \mathcal{O}(\Delta x^4)$. Therefore, since $\epsilon = \Delta x^r$ with $0 \leq r \leq 4$, it follows that

$$\sigma = \sum_{k=0}^2 \tilde{\alpha}_k = \sum_{k=0}^2 \gamma_k \left[1 + \left(\frac{|\mathbb{D}_{S_-}^4 f_{j+1/2}|^2}{\beta_k^2 + \epsilon} \right) \right] = 1 + \mathcal{O}(\Delta x^4),$$

the last equality being a consequence of the fact $\sum_{k=0}^2 \gamma_k = 1$. In view of this observation, a direct computation yields

$$\tilde{\omega}_k = \frac{\tilde{\alpha}_k}{\sigma} = \gamma_k(1 + O(\Delta x^4)). \quad (47)$$

With this estimate at hand, we will verify the relation in (46). For this, recall that $\sum_{k=0}^2 \gamma_k = 1$ and $\sum_{k=0}^2 \tilde{\omega}_k = 1$. Then, applying (39) and (45), we can derive the identity

$$\Delta x^2 \widehat{f''}_{j+\frac{1}{2}}^- - \mathbb{D}_{S_-}^2 f_{j+\frac{1}{2}} = \sum_{k=0}^2 (\tilde{\omega}_k - \gamma_k) (\mathbb{D}_{S_k}^2 f_{j+\frac{1}{2}} - \Delta x^2 f''_{j+\frac{1}{2}}).$$

As shown in (47), $\tilde{\omega}_k = \gamma_k + O(\Delta x^4)$. Also, $\mathbb{D}_{S_k}^2 f_{j+\frac{1}{2}} = f''_{j+\frac{1}{2}} \Delta x^2 + O(\Delta x^3)$. Therefore, it follows that

$$\Delta x^2 \widehat{f''}_{j+\frac{1}{2}}^- - \mathbb{D}_{S_-}^2 f_{j+\frac{1}{2}} = O(\Delta x^6).$$

In a similar way, we can get the relation for $\widehat{f''}_{j+\frac{1}{2}}^+$. It leads to the conclusion of this theorem. $\square$

The overall algorithm for the approximation to the second derivative of f is summarized below.

Algorithm: Approximation to $\frac{\partial^2 f}{\partial x^2}$ at $x = x_{j+1/2}$.

Input. The values f_n on S_- and the coefficients $\gamma_0 := -\frac{5}{24}$, $\gamma_1 := \frac{22}{24}$ and $\gamma_2 := \frac{7}{24}$.

Initialize. The second-order differences of f on S_k , i.e., $\mathbb{D}_{S_k}^2 f$ ($k = 0, 1, 2$) at $x = x_{j+1/2}$.

1. Evaluate the smoothness indicator $\beta_{f,k}$ on each substencil S_k .
2. Determine the value ζ in (42).
3. Compute the fourth-order central difference $\mathbb{D}_{S_-}^4 u$ and the unnormalize weights $\tilde{\alpha}_k$.
4. Using ζ and $\tilde{\alpha}_k$, compute the nonlinear weights $\tilde{\omega}_k$.
5. Define the approximation to f'' at $x = x_{j+1/2}$ as follows:

$$\Delta x^2 \widehat{f''}_{j+\frac{1}{2}}^- = \sum_{k=0}^2 \tilde{\omega}_k \mathbb{D}_{S_k}^2 f_{j+\frac{1}{2}}.$$

In a mirror symmetric way, we can construct $\Delta x^2 \widehat{f''}_{j+\frac{1}{2}}^+$ as in (45) with the given data values f_n on the stencil S_+ and the same coefficients γ_k ($k = 0, 1, 2$).

4 Numerical Results

In this section, we simulate some benchmark test problems of the one- and two-dimensional Euler equations to demonstrate the effectiveness of the proposed AWENO-E scheme. Our numerical experiments treat the issues of convergence order, computation cost, and ENO property of the final reconstruction. The two-dimensional system of Euler equations for gas dynamics is of the form

$$\mathbf{q}_t + \mathbf{F}_x + \mathbf{G}_y = 0 \quad (48)$$

with

$$\mathbf{q} = (\rho, \rho u, \rho v, E)^T, \quad \mathbf{F} = (\rho u, \rho u^2 + p, \rho uv, (E + p)u)^T, \\ \mathbf{G} = (\rho v, \rho vu, \rho v^2 + p, (E + p)v)^T.$$

Here $\mathbf{q}$ is the conservative variable, $\mathbf{F}$ and $\mathbf{G}$ indicate the x - and y - directional fluxes respectively. The equation of state is given as

$$p = (\gamma - 1) \left(E - \frac{1}{2} \rho (u^2 + v^2) \right)$$

with γ the ratio of specific heats. We set $\gamma = 1.4$ unless stated otherwise. The ρ, u, v, p and E are the density, x, y -velocities, pressure and total energy respectively. The one-dimensional Euler system can be deduced from the two-dimensional problem by removing the extra degree of freedom; see (49) and (50). The fluxes used in our simulation are the LF (Lax–Fridrichs) and HLLC functions. The HLLC flux function is a modified version of Harten, Lax and van Leer flux that improves HLL Riemann solver by incorporating the contact surface into the wave pattern. The numerical results of AWENO-E with LF and HLLC solvers are compared with those of AWENO-JS and AWENO-Z. For simplicity, we use the notation AWENO- xx - yy to stand for the ‘ xx ’ scheme with the ‘ yy ’ numerical flux, i.e., xx =JS, Z, or E, and yy =LF or HLLC flux.

4.1 Tests of Accuracy

In this subsection, we test the accuracy of our AWENO scheme on the one- and two-dimensional Euler systems for gas dynamics with smooth solutions. In our experiments for time integration in solving the system of ODEs, the fourth-order Non-TVD Runge–Kutta method is used with $\Delta t = 0.5 \cdot \Delta x^{6/4}$.

Example 4.1 We first investigate the spatial accuracy of the AWENO-E method for solving the one-dimensional Euler system of equations with a smooth initial condition:

$$\mathbf{q}_t + \mathbf{F}(\mathbf{q})_x = 0$$

with

$$\mathbf{q} = (\rho, \rho u, E)^T, \quad \mathbf{F} = (\rho u, \rho u^2 + p, (E + p)u)^T, \quad (49)$$

where ρ, u, E and p are the density, velocity, energy and pressure respectively. The initial condition is given as

$$\mathbf{q}(x, 0) = (\rho, u, p) = (1 + 0.5 \sin(4\pi x), 1, 1), \quad x \in [-1, 1], \quad (50)$$

and periodic boundary conditions are used. This simulation has been done until $t = 2$. The exact solution is

$$\mathbf{q}(x, t) = (1 + 0.5 \sin(4\pi(x - t)), 1, 1).$$

The L_1 and L_∞ errors and approximation orders for the density profile are given in Table 1. We observe that the AWENO-E scheme achieves the enhanced convergence order, that is, ‘six’, which confirms the theoretical results.

Example 4.2 In this example, we test the accuracy of the AWENO-E scheme for the two-dimensional system of Euler equations in (48) with the initial condition

$$\mathbf{q}(x, y, 0) = (\rho, u, v, p) = (1 + 0.5 \sin(4\pi(x + y)), 1, -0.5, 1), \quad (x, y) \in [-1, 1] \times [-1, 1],$$

Table 1 The L_1 , L_∞ errors and convergence orders for the density profile ρ at $t = 2$ for 1D Euler equation

N	AWENO-JS-LF		AWENO-Z-LF		AWENO-E-LF	
	L_∞	L_1	L_∞	L_1	L_∞	L_1
20	4.8E-01(–)	3.1E-01(–)	4.5E-01(–)	2.9E-01(–)	5.1E-01(–)	3.5E-01(–)
40	1.4E-01(1.8)	9.1E-02(1.8)	4.2E-02(3.4)	2.7E-02(3.4)	2.9E-03(7.5)	1.8E-03(7.6)
80	7.3E-03(4.2)	5.2E-03(4.1)	1.3E-03(5.0)	8.2E-04(5.1)	4.6E-05(6.0)	2.1E-05(6.5)
160	3.0E-04(4.6)	1.8E-04(4.8)	3.9E-05(5.1)	2.4E-05(5.1)	5.0E-07(6.5)	3.0E-07(6.1)
320	1.0E-05(4.9)	5.7E-06(5.0)	1.2E-06(5.0)	7.5E-07(5.0)	6.9E-09(6.2)	4.4E-09(6.1)
640	3.2E-07(5.0)	1.8E-07(5.0)	3.7E-08(5.0)	2.3E-08(5.0)	1.0E-10(6.1)	6.4E-11(6.1)
N	AWENO-JS-HLLC		AWENO-Z-HLLC		AWENO-E-HLLC	
	L_∞	L_1	L_∞	L_1	L_∞	L_1
20	5.3E-01(–)	3.5E-01(–)	4.0E-01(–)	2.7E-01(–)	2.7E-01(–)	1.9E-01(–)
40	6.4E-02(3.1)	4.3E-02(3.0)	1.8E-02(4.5)	1.2E-02(4.5)	3.0E-03(6.5)	1.9E-03(6.6)
80	3.4E-03(4.3)	2.2E-03(4.3)	5.0E-04(5.2)	3.1E-04(5.3)	4.5E-05(6.1)	2.2E-05(6.4)
160	1.2E-04(4.8)	6.9E-05(5.0)	1.5E-05(5.1)	9.0E-06(5.1)	5.1E-07(6.5)	2.8E-07(6.3)
320	4.1E-06(4.9)	2.2E-06(5.0)	4.4E-07(5.0)	2.8E-07(5.0)	6.8E-09(6.2)	4.3E-09(6.0)
640	1.3E-07(5.0)	6.7E-08(5.0)	1.4E-08(5.0)	8.8E-09(5.0)	1.0E-10(6.1)	6.4E-11(6.1)

where periodic boundary conditions are used. The simulation was performed up to time $t = 2$. The L_1 and L_∞ approximation errors and convergence orders of several AWENO schemes for the density profile are presented in Table 2. The exact solution is

$$\mathbf{q}(x, t) = (1 + 0.5 \sin(4\pi(x + y - 0.5t)), 1, -0.5, 1).$$

We can clearly find that the AWENO-E scheme achieves the designed order of accuracy. The approximation errors computed by the AWENO-E scheme (with LF and HLLC solvers) are smaller than those by the AWENO-JS and AWENO-Z schemes. In particular, the L_1 errors obtained by the AWENO-E with HLLC solver are slightly smaller than those computed by AWENO-E with LF solver, especially on low-resolution grids.

Further, we compare the computational efficiency of the tested AWENO schemes in terms of the CPU time versus L_1 or L_∞ errors using different grids. In particular, the AWENO-E scheme is compared with the sixth-order AWENO-Z schemes (denoted by AWENO-Z7-LF, AWENO-Z7-HLLC), which have the seventh-order interpolation and the sixth-order flux correction methods. The graphs of CPU times against errors are displayed in Fig. 2. Each marker in these graphs depicts the pairs of CPU time and L_∞ -approximation errors at 10×2^k grid points with $k = 1, \dots, 6$. We can observe that AWENO-E outperforms the other AWENO schemes.

4.2 Tests of One-Dimensional Euler System

In this subsection, we present numerical results on the one-dimensional Euler system of gas dynamics. For all numerical tests, the CFL number is fixed as 0.5 with TVD Runge–Kutta third order scheme.

Table 2 The L_1 , L_∞ errors and convergence orders for the density profile ρ at $t = 2$ for 2D Euler equation

N	AWENO-JS-LF		AWENO-Z-LF		AWENO-E-LF	
	L_∞	L_1	L_∞	L_1	L_∞	L_1
10×10	4.8E-01(−)	3.1E-01(−)	4.5E-01(−)	2.9E-01(−)	4.9E-01(−)	3.1E-01(−)
20×20	2.1E-01(1.2)	1.4E-01(1.2)	6.9E-02(2.7)	4.5E-02(2.7)	5.9E-03(6.4)	4.3E-03(6.2)
40×40	1.2E-02(4.1)	9.0E-03(3.9)	2.3E-03(4.9)	1.5E-03(4.9)	3.8E-05(7.3)	1.9E-05(7.8)
80×80	5.0E-04(4.6)	3.4E-04(4.7)	7.0E-05(5.0)	4.3E-05(5.1)	2.8E-07(7.1)	1.6E-07(6.9)
160×160	1.8E-05(4.8)	1.0E-05(5.0)	2.2E-06(5.0)	1.4E-06(5.0)	3.5E-09(6.3)	2.2E-09(6.2)
320×320	5.7E-07(5.0)	3.3E-07(5.0)	6.7E-08(5.0)	4.2E-08(5.0)	5.4E-11(6.0)	3.4E-11(6.0)
AWENO-JS-HLLC						
10×10	4.7E-01(−)	3.0E-01(−)	3.5E-01(−)	2.2E-01(−)	3.2E-01(−)	2.0E-01(−)
20×20	8.3E-02(2.5)	5.6E-02(2.4)	2.4E-02(3.9)	1.6E-02(3.8)	3.1E-03(6.7)	1.3E-03(7.3)
40×40	4.4E-03(4.2)	3.1E-03(4.2)	7.2E-04(5.1)	4.6E-04(5.1)	2.1E-05(7.2)	8.9E-06(7.2)
80×80	1.7E-04(4.6)	1.0E-04(4.9)	2.2E-05(5.1)	1.3E-05(5.1)	2.7E-07(6.3)	1.4E-07(6.0)
160×160	5.8E-06(4.9)	3.2E-06(5.0)	6.6E-07(5.0)	4.2E-07(5.0)	3.4E-09(6.3)	2.1E-09(6.0)
320×320	1.8E-07(5.0)	1.0E-07(5.0)	2.1E-08(5.0)	1.3E-08(5.0)	5.4E-11(6.0)	3.4E-11(6.0)
AWENO-Z-HLLC						
10×10	4.7E-01(−)	3.0E-01(−)	3.5E-01(−)	2.2E-01(−)	3.2E-01(−)	2.0E-01(−)
20×20	8.3E-02(2.5)	5.6E-02(2.4)	2.4E-02(3.9)	1.6E-02(3.8)	3.1E-03(6.7)	1.3E-03(7.3)
40×40	4.4E-03(4.2)	3.1E-03(4.2)	7.2E-04(5.1)	4.6E-04(5.1)	2.1E-05(7.2)	8.9E-06(7.2)
80×80	1.7E-04(4.6)	1.0E-04(4.9)	2.2E-05(5.1)	1.3E-05(5.1)	2.7E-07(6.3)	1.4E-07(6.0)
160×160	5.8E-06(4.9)	3.2E-06(5.0)	6.6E-07(5.0)	4.2E-07(5.0)	3.4E-09(6.3)	2.1E-09(6.0)
320×320	1.8E-07(5.0)	1.0E-07(5.0)	2.1E-08(5.0)	1.3E-08(5.0)	5.4E-11(6.0)	3.4E-11(6.0)

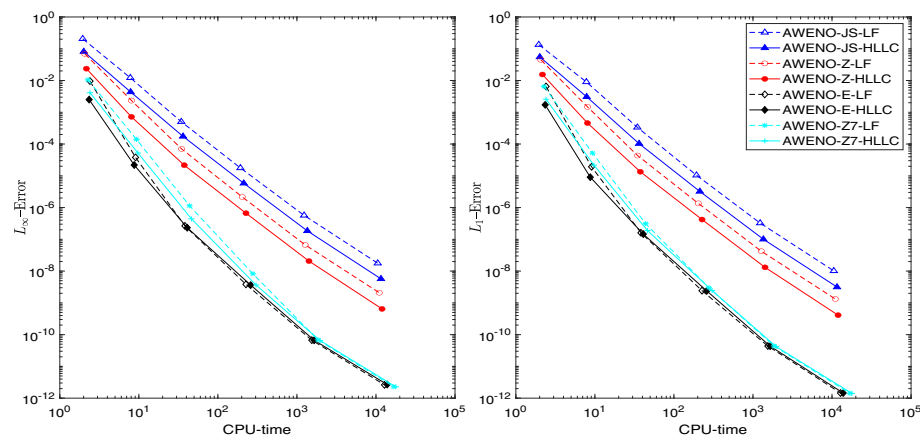


Fig. 2 Computational efficiency (CPU time versus errors) of AWENO-JS, AWENO-Z, and AWENO-E with LF and HLLC flux functions for two-dimensional system of Euler Eq. (48)

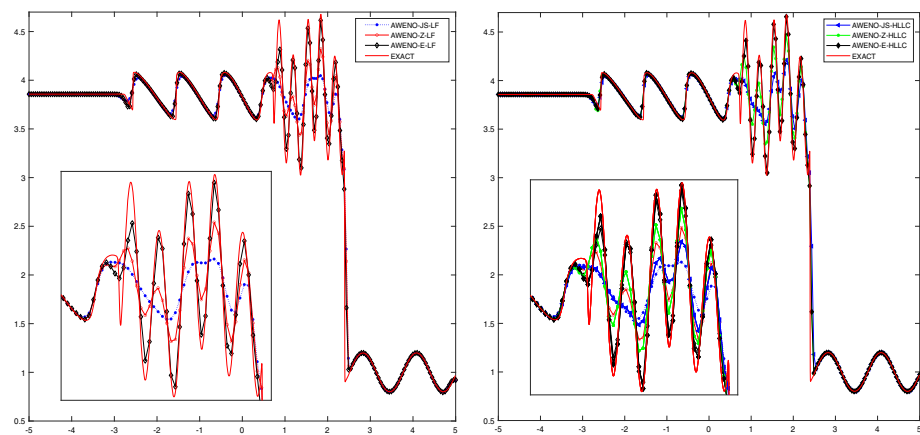


Fig. 3 Density profiles of the shock-entropy interacting of Shu–Osher [49] obtained by AWENO-JS, AWENO-Z and AWENO-E with LF and HLLC solvers at $t = 1.8$ and $\Delta x = 1/20$

Example 4.3 (Shock-density wave interaction) In this example, we are concerned with the so-called ‘Shu–Osher problem’ [49] with the following initial condition:

$$(\rho, u, p) = \begin{cases} (3.857143, 2.629369, 10.33333) & \text{for } x \in [-5, -4), \\ (1 + 0.2 \sin(5x), 0, 1.0) & \text{for } x \in [-4, 5]. \end{cases}$$

It represents a Mach 3 shock-density wave interacting with a sine entropy wave carrying density fluctuations. Since entropy waves are damped at high numerical dissipation, this test case allows us to assess the effect of nonlinear viscosity and diffusivity on turbulence. A numerical solution of the fifth-order WENO-JS method using 3200 grids is chosen as the reference solution. The computed density profiles ρ using several AWENO schemes with $\Delta x = 1/20$ at $t = 1.8$ are represented in Fig. 3. It can be observed that the proposed AWENO-E method shows better resolution than the AWENO-JS and AWENO-Z schemes in resolving

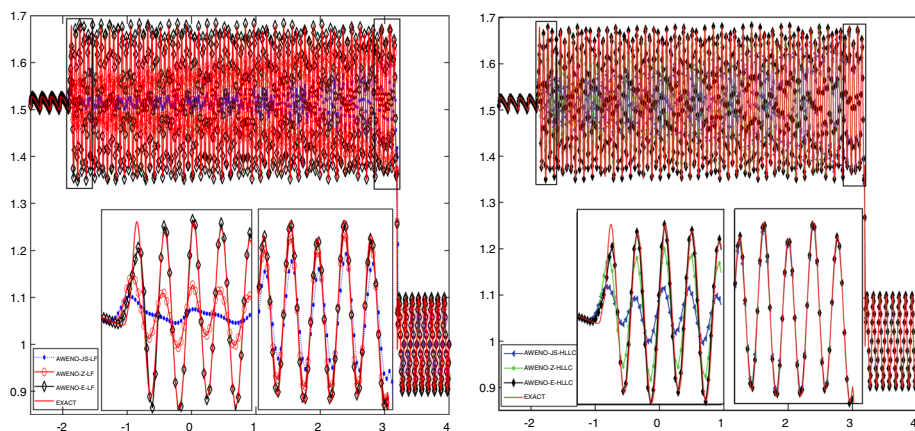


Fig. 4 Density profiles of the Titarev–Toro problem [51] obtained by AWENO-JS, AWENO-Z and AWENO-E with LF and HLLC solvers at $t = 5$ and $\Delta x = 1/150$

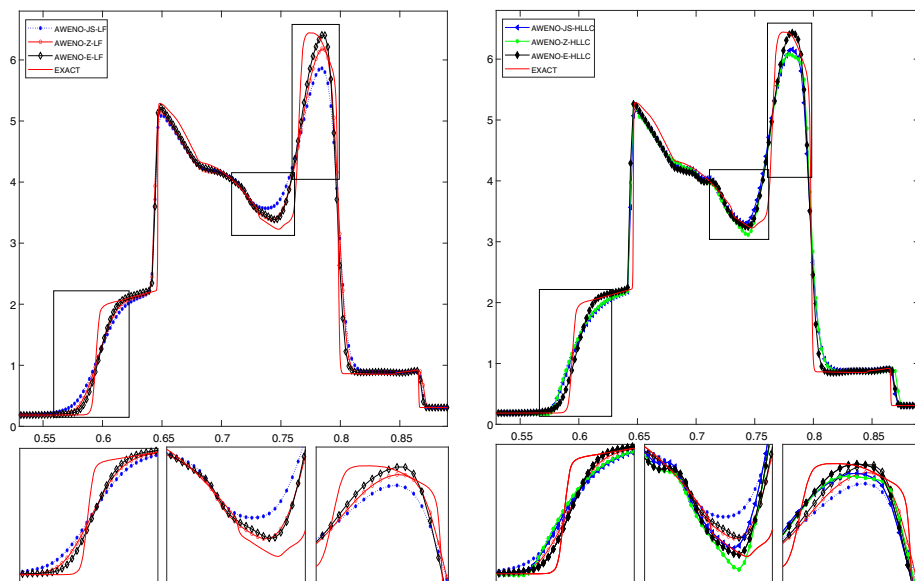


Fig. 5 Density profiles of the interacting blast wave problem [56] obtained by AWENO-JS, AWENO-Z and AWENO-E with LF and HLLC fluxes at $t = 0.038$ and $\Delta x = 1/400$

the small structures downstream of the shock wave. In particular, the AWENO-E-HLLC scheme is superior to other tested methods in resolving wave crests and troughs.

Furthermore, we simulate the ‘Titarev–Toro’ problem with the following initial condition [51]

$$(\rho, u, p) = \begin{cases} (1.515695, 0.523346, 1.805000) & x \in [-5, -4), \\ (1 + 0.1 \sin(20\pi x), 0, 1) & x \in [-4, 5], \end{cases}$$

which is a variation of the Shu–Osher problem. This test is computed with $\Delta x = 1/150$ until $t = 5$. The reference solution is obtained by the WENO-JS scheme with 5000 grid

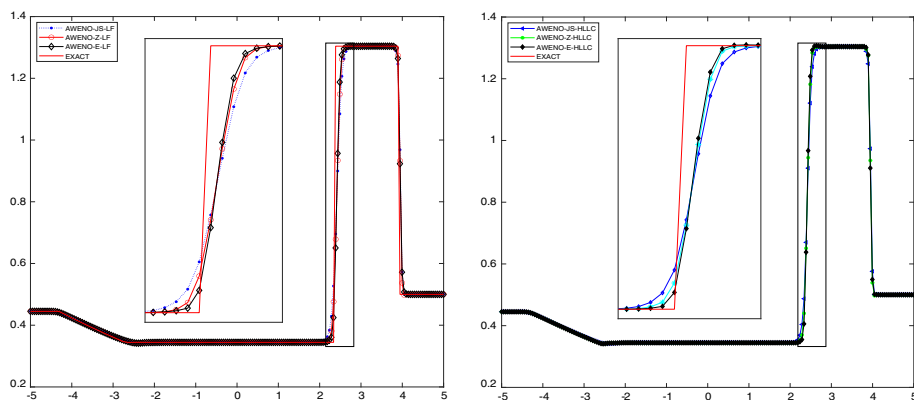


Fig. 6 Numerical results of Lax problem [34] obtained by AWENO-JS, AWENO-Z and AWENO-E with LF and HLLC solvers at $t = 1.6$ and $\Delta x = 1/20$

points. The numerical results are displayed in Fig. 4. We see that the high frequency entropy waves behind the shock obtained by AWENO-E with LF and HLLC solvers using 1500 grid points are very similar to the reference solution. The oscillatory wave patterns after shock entropy wave interactions are better captured by our proposed AWENO-E method than other AWENO-JS and AWENO-Z methods.

Example 4.4 (Interacting blast wave problem) We test the two strong shocks interaction case which is called the ‘blast wave problem’ proposed by Woodward and Colella [56]. This is characterized by the strong shock waves, which can be used to validate the shock-capturing capability of a scheme. The initial condition of this problem is formulated as

$$(\rho, u, p) = \begin{cases} (1, 0, 1000), & x \in [0, 0.1], \\ (1, 0, 0.01), & x \in [0.1, 0.9], \\ (1, 0, 100), & x \in [0.9, 1], \end{cases}$$

where reflective boundary condition is applied at the both end points of the domain $[0, 1]$. A highly resolved solution of the fifth order WENO-JS scheme [28] on 3200 grid points is set as a reference solution. The simulation was performed until time $t = 0.038$ with $\Delta x = 1/400$. The density profile ρ computed by the AWENO-E, AWENO-JS and AWENO-Z schemes with LF and HLLC fluxes are plotted in Fig. 5 with $\Delta x = 1/400$ at time $t = 0.038$. It can be noticed that all of the results agree well with the exact density profiles and the AWENO-E-HLLC scheme achieves the highest resolution in resolving the peak value of the density profile at $x = 0.78$.

Example 4.5 (Lax problem) We now test the ‘Lax problem’ [34]. The initial conditions for density, velocity, and pressure are given as follows:

$$(\rho, u, p) = \begin{cases} (0.445, 0.698, 3.528) & \text{for } x \in [-5, 0), \\ (0.5, 0, 0.571) & \text{for } x \in [0, 5]. \end{cases}$$

We calculate the solution on $\Delta x = 10/200$ to time $t = 1.6$. Figure 6 gives a comparison of the exact density distribution (reported in Toro [52]) and the computed density profiles. We observe that the results of AWENO-E agree well with the reference solution. Moreover,

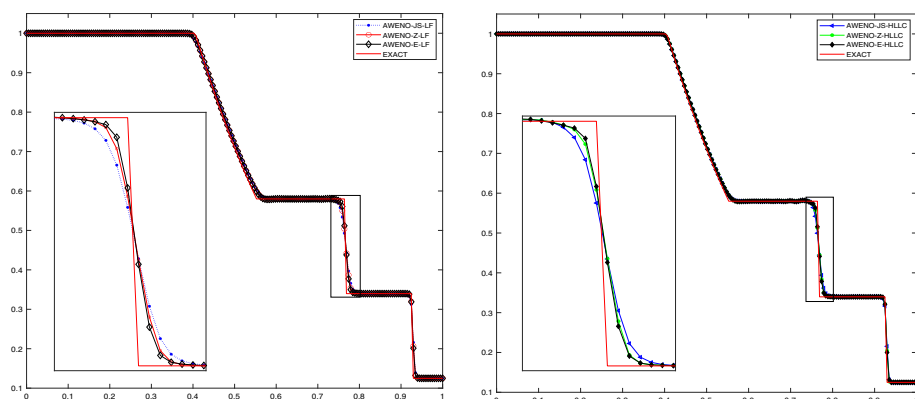


Fig. 7 Numerical results of Sod problem [50] obtained by AWENO-JS, AWENO-Z, and AWENO-E with LF and HLLC solvers at $t = 0.2$ and $\Delta x = 1/200$

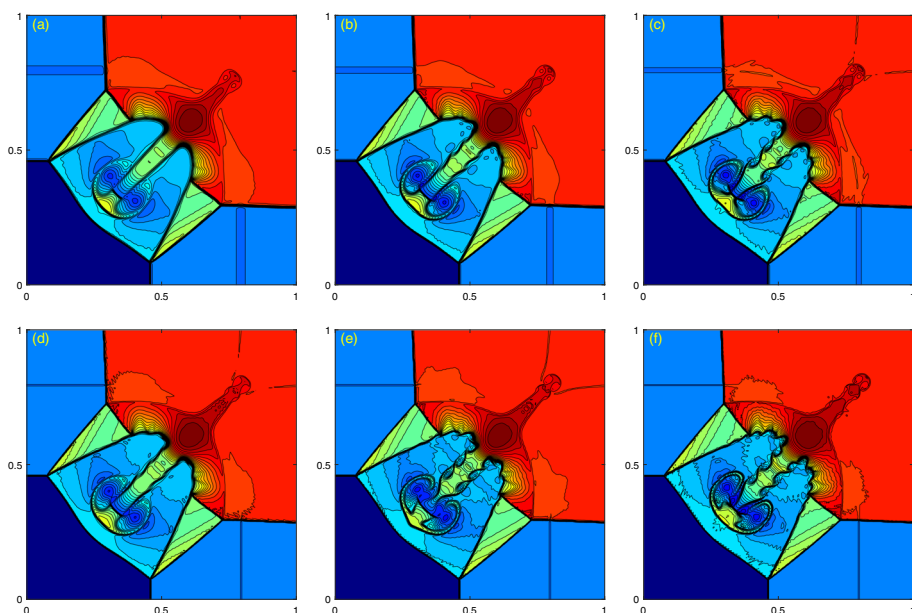


Fig. 8 Density profiles of 2D Riemann Problem [44] at $t = 0.8$ and $\Delta x = \Delta y = 1/300$: **a** AWENO-JS, **b** AWENO-Z, **c** AWENO-E with LF flux and **d** AWENO-JS, **e** AWENO-Z, **f** AWENO-E with HLLC flux

among all the AWENO schemes, the contact wave is better captured by the AWENO-E with HLLC solver.

Example 4.6 (Sod problem) We consider the ‘Sod shock tube problem’ [50]. The computational domain is $[0, 1]$, which is divided into equally-spaced 200 cells. The initial conditions for density, velocity, and pressure are given as follows:

$$(\rho, u, p) = \begin{cases} (1, 0, 1) & \text{for } x \in [0, 0.5), \\ (0.125, 0, 0.1) & \text{for } x \in [0.5, 1]. \end{cases}$$

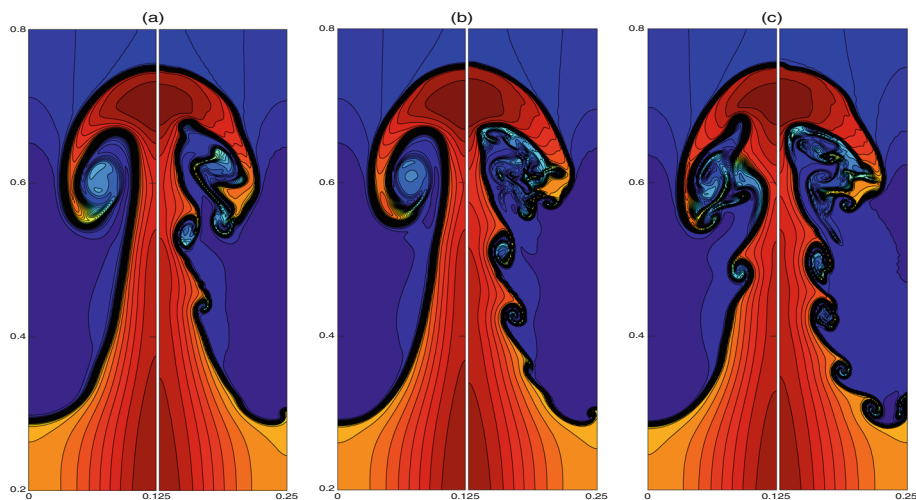


Fig. 9 Density profiles of Rayleigh-Taylor instability problem at $t = 1.95$ and $\Delta x = \Delta y = 1/600$: **a** AWENO-JS, **b** AWENO-Z, and **c** AWENO-E with LF flux (left) and HLLC flux (right)

The density profile ρ is plotted in Fig. 7 at $t = 0.2$. The exact solution is obtained from the exact Riemann solver that is reported by Toro [52]. A good agreement between the numerical results and the exact solution is observed, and the propagation of the shock wave is well captured by all of the AWENO schemes. The AWENO-E-HLLC scheme produces better results at the contact wave than the other tested AWENO schemes.

4.3 Tests of Two-Dimensional Euler System

In what follows, we now present numerical results on the two-dimensional Euler system of gas dynamics. For all numerical tests, the CFL number is fixed as 0.2 with third order TVD Runge-Kutta scheme.

Example 4.7 (Two-dimensional Riemann problem-configuration 3) We consider the well-known configuration 3 of the two-dimensional Riemann problem proposed in [44]. The spatial domain is square $[0, 1] \times [0, 1]$ and divided into four quadrants by the lines $x = 0.8$ and $y = 0.8$. In each quadrant, the initial data is defined by a constant and evolved until time $t = 0.8$. Specifically, the initial condition in each quadrant is defined as:

$$(\rho, u, v, p) = \begin{cases} (1.5, 0, 0, 1.5) & \text{if } (x, y) \in [0.8, 1] \times [0.8, 1], \\ (0.5323, 1.206, 0, 0.3) & \text{if } (x, y) \in [0, 0.8] \times [0.8, 1], \\ (0.138, 1.206, 1.206, 0.029) & \text{if } (x, y) \in [0, 0.8] \times [0, 0.8], \\ (0.5323, 0, 1.206, 0.3) & \text{if } (x, y) \in [0.8, 1] \times [0, 0.8], \end{cases}$$

with outflow boundary conditions. The density contours of numerical results obtained by applying the AWENO schemes on 300×300 grid points (at $t = 0.8$) are plotted in Fig. 8. The results of AWENO-E with LF and HLLC solvers are compared with those of AWENO-JS and AWENO-Z. Clearly, AWENO-E presents more vortices along contact lines than the AWENO-JS and AWENO-Z schemes. In particular, AWENO-E with HLLC solver captures fine scale structures with high resolution better than the other AWENO schemes.

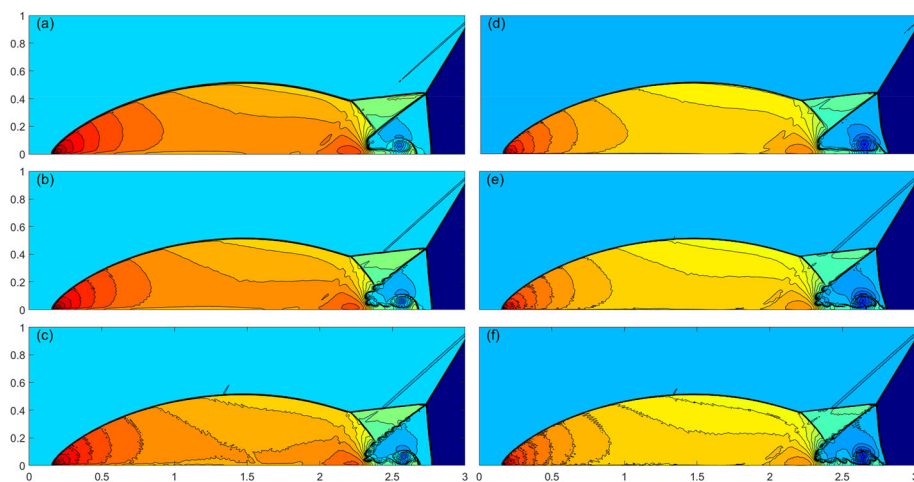


Fig. 10 Density profiles of Double Mach reflection of a strong shock [56] at $t = 0.2$ with $\Delta x = \Delta y = 1/240$: **a** AWENO-JS, **b** AWENO-Z, and **c** AWENO-E with LF flux and **d** AWENO-JS, **e** AWENO-Z, and **f** AWENO-E with HLLC flux

Example 4.8 (Two-dimensional Rayleigh-Taylor instability) The so-called RTI problem has been simulated to identify differences in the numerical dissipation of various higher-order accuracy methods (e.g., see [15, 46, 57, 58]). The initial condition is given as

$$(\rho, u, v, p) = \begin{cases} (2, 0, -0.025\sqrt{\frac{5p}{3\rho}}\cos(8\pi x), 2y + 1), & \text{for } y \in [0, 0.5), \\ (1, 0, -0.025\sqrt{\frac{5p}{3\rho}}\cos(8\pi x), y + 1.5), & \text{for } y \in [0.5, 1], \end{cases}$$

on the spatial domain $[0, 0.25] \times [0, 1]$. This problem describes the phenomenon that a heavier fluid superimposes on a lighter fluid in a gravitational field where the fluid interface is unstable. Any perturbation at this interface typically evolves over time, indicating the penetration of heavier and lighter fluids into each other. The gravitational effect can be obtained by adding ρ and ρv to the right of y -momentum and the energy equation respectively. The ratio of specific heats is $\gamma = 5/3$. The right and left boundary are taken by reflective boundary conditions and $(\rho, u, v, p) = (1, 0, 0, 2.5)$ for top and $(\rho, u, v, p) = (2, 0, 0, 1)$ for bottom boundary conditions. The computational domain is discretized with 150×600 grid points and the simulation is performed until time $t = 1.95$. Figure 9 displays the density contours of the numerical results of the simulated AWENO schemes. We can see that AWENO-E gives significantly improved resolutions to the AWENO-JS and AWENO-Z schemes. In particular, the AWENO-E scheme with HLLC flux captures more vortices in density than the other cases.

Example 4.9 (Double Mach Reflection problem) We simulate the two-dimensional double Mach reflection (DMR) problem that was originally considered by Woodward and Colella [56]. It involves a vertical shock wave moving horizontally into a wedge inclined by some angle. When the shock wave hits the sloping wall, a complicated flow structure occurs,

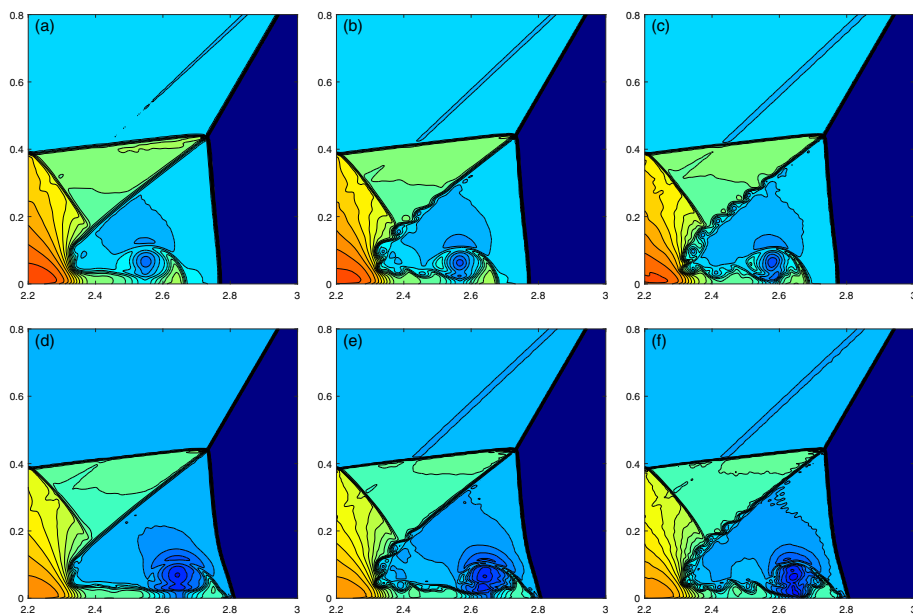


Fig. 11 Density profiles of zoomed region $[2.2, 3] \times [0, 0.8]$ for double mach reflection problem at $t = 0.2$ and $\Delta x = \Delta y = 1/240$: **a** AWENO-JS, **b** AWENO-Z, and **c** AWENO-E with LF flux and **d** AWENO-JS, **e** AWENO-Z, and **f** AWENO-E with HLLC flux

involving multiple shock reflections and slip lines. The initial condition is given by

$$(\rho, u, v, p) = \begin{cases} (1.4, 0, 0, 1.0), & \text{if } x > 1/6 + \sqrt{3}/3y, \\ (8.0, 7.1447, -4.125, 116.5), & \text{otherwise,} \end{cases}$$

on the spatial domain $[0, 4] \times [0, 1]$. Initially, a right-moving Mach 10 shock wave is placed at $(x, y) = (\frac{1}{6}, 0)$ with an incidence angle of 60° to the horizontal axis. For the bottom boundary, an exact post-shock condition is positioned on $x \in [0, \frac{1}{6}]$, and the rest part of the bottom is used as a reflective boundary condition. Left and right boundaries use inflow and outflow boundary conditions respectively. For the top boundary condition, the exact state values of the Mach 10 shock are used in a time dependent manner. The simulation was performed until time $t = 0.2$ and the blow-up region around the double Mach stems. The numerical results of AWENO-E with LF and HLLC solvers with 960×240 grid points are compared with the results of AWENO-JS and AWENO-Z in Fig. 10. To facilitate our comparison, the density contours zoomed around the double Mach reflection are displayed in Fig. 11. Compared to the cases of AWENO-JS and AWENO-Z, it can be seen that AWENO-E provides an improved resolution at the contact discontinuities. In particular, the complex features of the flow are well-captured by AWENO-E.

Example 4.10 (Mach 3 wind tunnel with a forward-facing step problem) In this example, we simulate the well-known benchmark problem of the wind tunnel with a forward-facing step onto which gas flows at Mach 3 [11, 56]. The setup of this simulation is given as follows: The wind tunnel is 1 length unit width and 3 length unit long. The step is located 0.2 length units high and 0.6 length units from the left end of the tunnel. A gas with a density of 1.4, a pressure of 1.0 and a velocity of 3 is continuously supplied at the left boundary. Reflective

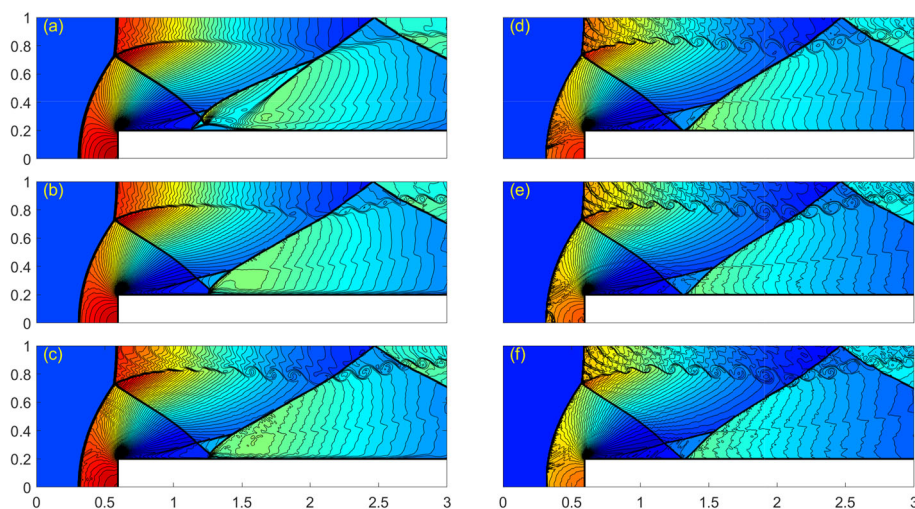


Fig. 12 Density profiles of wind tunnel problem at $t = 4$ and $\Delta x = \Delta y = 1/200$: **a** AWENO-JS, **b** AWENO-Z, and **c** AWENO-E with LF flux and **d** AWENO-JS, **e** AWENO-Z, and **f** AWENO-E with HLLC flux

boundary conditions are applied along the walls of the tunnel. Inflow and outflow boundary conditions are applied at the entrance and the exit respectively. Due to the interactions and reflections of the shocks, a triple point is formed from which the trail of vortices moves towards the right boundary. The simulation is performed up to $t = 4$ with 600×200 grids. The density profiles obtained by AWENO-E with the other AWENO schemes are displayed in Fig. 12. We clearly observe that the AWENO-E with LF and HLLC solvers could offer better results than the other AWENO schemes. In particular, the roll-up of the vortex sheet is more clearly visible with AWENO-E.

Example 4.11 (Explosion) This explosion problem, introduced in [52], is a radially symmetric two-dimensional problem with higher density and pressure inside an initial circular area. The circle is assumed to be centered at the origin with radius $r = 0.4$. The spatial domain is $[-1.5, 1.5] \times [-1.5, 1.5]$ and discretized with 800×800 grid points. We simulate this problem up to time $t = 3.2$ with the initial condition

$$(\rho, u, v, p) = \begin{cases} (1.000, 0, 0, 1.0) & \text{if } x^2 + y^2 < 0.4^2, \\ (0.125, 0, 0, 0.1) & \text{otherwise.} \end{cases}$$

In Fig. 13, the numerical results of AWENO-E with LF and HLLC solvers are compared with those of AWENO-JS, AWENO-Z, and AWENO-E. Here, to ease our comparison, the first quadrants of the density profiles are displayed. It can be seen that the numerical result computed by AWENO-E with HLLC solver is more “curlier” at the contact surface than the other results. It means that the AWENO-E-HLLC is less dissipative.

5 Conclusion

In this study, we introduced a new formulation of an alternative AWENO scheme improving the performance of the known fifth-order AWENO schemes. To enhance the order of accuracy,

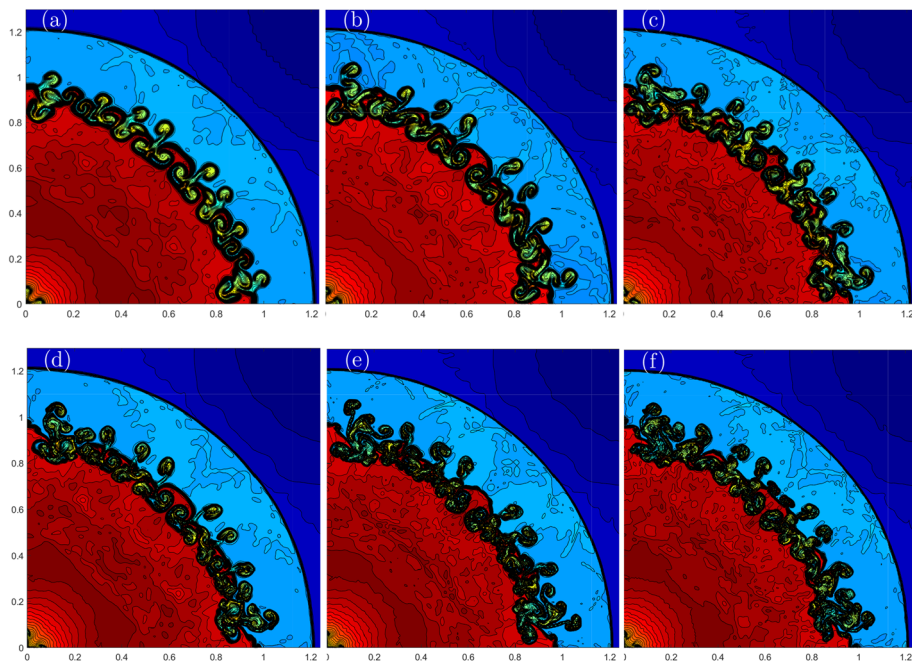


Fig. 13 Density profiles on the first quadrant for the Explosion problem [52] at $t = 3.2$ and $\Delta x = \Delta y = 3/800$: **a** AWENO-JS, **b** AWENO-Z, and **c** AWENO-E with LF flux and **d** AWENO-JS, **e** AWENO-Z, and **f** AWENO-E with HLLC flux

the low-order numerical flux is implemented by a WENO interpolator based on exponential polynomials with a shape parameter. By choosing a locally fitting shape parameter, the proposed AWENO attains the overall sixth-order accuracy in smooth regions under the same fifth-order AWENO framework, while maintaining the essentially non-oscillatory behavior around discontinuities. Furthermore, we present a new strategy of limiting procedure to deal with the second-order derivative term in the high-order correction flux to avoid generating redundant oscillations near strong shocks. The efficiency of the new AWENO method was confirmed by some experimental results. In the future, we would like to extend this work to higher-order alternative WENO schemes.

6 Appendix

- *The Lax–Friedrichs (LF) flux.* The LF flux is defined by

$$\hat{f}_{j+1/2}^{\text{LF}}(u^-, u^+) = \frac{1}{2}[(f(u^-) + f(u^+)) - \alpha(u^+ - u^-)],$$

where α is taken as an upper bound over the whole line for $|f'(u)|$ in the scalar case, or the absolute value of eigenvalues of the Jacobian for the system case.

- *The HLLC flux.* The HLLC flux is a modified version of the HLL flux, whereby the missing contact and shear waves are restored. The HLLC flux for the Euler equations

$q = (\rho, \rho u, E)^T$ is given by

$$\hat{f}_{j+1/2}^{\text{HLLC}}(q^-, q^+) = \begin{cases} f(q^-) & \text{if } 0 \leq s^-, \\ f(q^-) + s^-(q^{0-} - q^-) & \text{if } s^- \leq s^0, \\ f(q^+) + s^+(q^{0+} - q^+) & \text{if } s^0 \leq s^+, \\ f(q^+) & \text{if } s^+ \leq 0, \end{cases} \quad (51)$$

where by defining the averaging operation $\bar{f} = \frac{1}{2}(f^+ + f^-)$ and the difference operation $\Delta f = f^+ - f^-$,

$$q^{0\pm} = \rho^\pm \frac{s^\pm - u^\pm}{s^\pm - s^0} \left[\frac{1}{s^0} \left(\frac{E^\pm}{\rho^\pm} + (s^0 - u^\pm) \left(s^0 + \frac{p^\pm}{s^\pm - u^\pm} \right) \right) \right],$$

$$p^0 = \bar{p} - \frac{1}{2} \Delta \bar{\rho} \bar{c}, \quad s^0 = \bar{u} - \frac{1}{2} \frac{\Delta p}{\bar{\rho} \bar{c}}, \quad s^\pm = u^\pm \pm c^\pm Q^\pm,$$

$$Q^\pm = \begin{cases} 1 & \text{if } p^0 \leq p^\pm, \\ \left(1 + \frac{\gamma+1}{2\gamma} \left(\frac{p^0}{p^\pm} - 1 \right) \right)^{1/2} & \text{if } p^\pm \leq p^0. \end{cases}$$

- **Ideal weights d_k based on exponential polynomials.** Let $\bar{x} = x_{j+1/2}$. The explicit forms of the Lagrange functions based on exponential polynomials in (16) are given as follows:

$$\begin{aligned} \ell_0(\bar{x}) &= \frac{3}{128} - \frac{5}{1024}(\lambda \Delta x)^2 + \frac{65}{98304}(\lambda \Delta x)^4 - \frac{179}{2359296}(\lambda \Delta x)^6 + \mathcal{O}(\Delta x^8) \\ \ell_1(\bar{x}) &= -\frac{5}{32} + \frac{(\lambda \Delta x)^2}{128} - \frac{29}{24576}(\lambda \Delta x)^4 + \frac{53}{368640}(\lambda \Delta x)^6 + \mathcal{O}(\Delta x^8) \\ \ell_2(\bar{x}) &= \frac{45}{64} + \frac{3}{512}(\lambda \Delta x)^2 - \frac{7}{16384}(\lambda \Delta x)^4 + \frac{47}{1966080}(\lambda \Delta x)^6 + \mathcal{O}(\Delta x^8) \\ \ell_3(\bar{x}) &= \frac{15}{32} - \frac{1}{64}(\lambda \Delta x)^2 + \frac{43}{24576}(\lambda \Delta x)^4 - \frac{259}{1474560}(\lambda \Delta x)^6 + \mathcal{O}(\Delta x^8) \\ \ell_4(\bar{x}) &= -\frac{5}{128} + \frac{7}{1024}(\lambda \Delta x)^2 - \frac{79}{98304}(\lambda \Delta x)^4 + \frac{989}{11796480}(\lambda \Delta x)^6 + \mathcal{O}(\Delta x^8) \end{aligned} \quad (52)$$

Moreover, letting $\bar{c}_\lambda = \cosh(\frac{\lambda \Delta x}{2})$, the linear weights $\bar{d}_k$ based on exponential polynomials are of the form:

$$\begin{aligned} \bar{d}_0 &= \frac{\bar{c}_\lambda + 2}{12\bar{c}_\lambda(\bar{c}_\lambda + 1)^2} = \frac{1}{16} - \frac{5}{384}(\lambda \Delta x)^2 + \mathcal{O}(\Delta x^4) \\ \bar{d}_1 &= \frac{6\bar{c}_\lambda^3 + 12\bar{c}_\lambda^2 + \bar{c}_\lambda - 4}{6\bar{c}_\lambda(\bar{c}_\lambda + 1)^2} = \frac{5}{8} + \frac{13}{192}(\lambda \Delta x)^2 + \mathcal{O}(\Delta x^4) \\ \bar{d}_2 &= \frac{3\bar{c}_\lambda + 2}{4\bar{c}_\lambda(\bar{c}_\lambda + 1)^2} = \frac{5}{16} - \frac{7}{128}(\lambda \Delta x)^2 + \mathcal{O}(\Delta x^4). \end{aligned} \quad (53)$$

- **Limiters for the correction term $\hat{f}^H$.** To demonstrate the advantage of our limiters to the second derivative in the correction term $\hat{f}^H$, we provide a comparison of performance between AWENO schemes with limited second derivative and central differences. In Fig. 14, we display the numerical results of the tested AWENO-E schemes for ‘Mach 3 wind tunnel problem’ in Example 4.10. We can see that the AWENO-E scheme with central differences generates more oscillations.

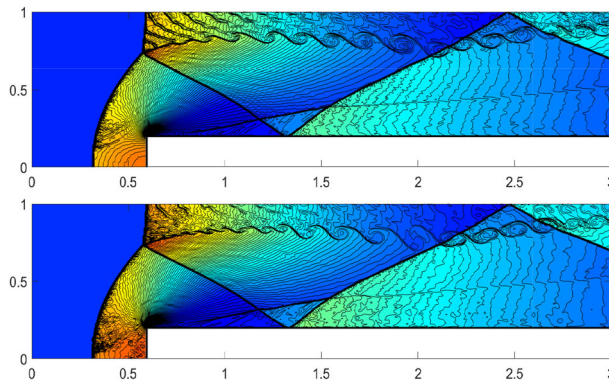


Fig. 14 Density profiles of wind tunnel problem at $t = 4$: (top) AWENO-E with limiters to the second derivative. (bottom) AWENO-E with the second-order central difference

Acknowledgements This research was supported by the grants NRF-2022R1F1A1066389 (H. Yang), and RS-2023-00208864, NRF-2019R1A6A1A11051177 (J. Yoon) of the National Research Foundation of Korea.

Funding Youngsoo Ha were supported by the grant NRF-2021R1A2C1095443 through the National Research Foundation of Korea, Hyoseon Yang is supported by NRF-2022R1F1A1066389 and Jungho Yoon is supported by RS-2023-00208864 and NRF-2019R1A6A1A11051177.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Acker, F., Borges, R., Costa, B.: An improved WENO-Z scheme. *J. Comput. Phys.* **313**, 726–753 (2016)
2. Balsara, D.S., Bhoriya, D., Shu, C.-W., Kumar, H.: Efficient alternative finite difference WENO schemes for hyperbolic systems with non-conservative products. <https://doi.org/10.48550/arXiv.2403.01266> (2024)
3. Balsara, D.S., Shu, C.-W.: Monotonicity preserving WENO schemes with increasingly high-order of accuracy. *J. Comput. Phys.* **160**, 405–452 (2000)
4. Balsara, D.S., Garain, S., Shu, C.-W.: An efficient class of WENO schemes with adaptive order. *J. Comput. Phys.* **326**, 780–804 (2016)
5. Bigoni, C., Hesthaven, J.: Adaptive WENO methods based on radial basis function reconstruction. *J. Sci. Comput.* **72**, 986–1020 (2017)
6. Borges, R., Carmona, M., Costa, B., Don, W.S.: An improved WENO scheme for hyperbolic conservation laws. *J. Comput. Phys.* **227**, 3191–3211 (2008)
7. Chen, L.L., Huang, C.: An improved WLS-WENO method for solving hyperbolic conservation laws. *J. Comput. Phys.* **392**, 96–114 (2019)
8. Christlieb, A.J., Feng, X., Jiang, Y., Tang, Q.: A high-order finite difference WENO scheme for ideal magnetohydrodynamics on curvilinear meshes. *SIAM J. Sci. Comput.* **40**(4), A2631–A2666 (2018)
9. Cravero, I., Semplice, M.: On the accuracy of WENO and CWENO reconstructions of third order on nonuniform meshes. *J. Sci. Comput.* **67**(3), 1219–1246 (2016)
10. Don, W.S., Li, D.-M., Gao, Z., Wang, B.-S.: A characteristic-wise alternative WENO-Z finite difference scheme for solving the compressible multicomponent non-reactive flows in the overestimated quasi-conservative form. *J. Sci. Comput.* **82**, 27 (2020)

11. Emery, A.F.: An evaluation of several differencing methods for inviscid fluid flow problems. *J. Comput. Phys.* **2**, 306–331 (1968)
12. Gao, Z., Fang, L., Wang, B.-S., Wang, Y., Don, W.S.: Seventh and ninth orders characteristic-wise alternative WENO finite difference schemes for hyperbolic conservation laws. *Comput. Fluids* **202**, 104519 (2020)
13. Gottlieb, S., Mullen, J.S., Ruuth, S.J.: A fifth order flux implicit WENO method. *J. Sci. Comput.* **27**, 271–287 (2006)
14. Gerolymos, G.A., Sénéchal, D., Vallet, I.: Very-high-order WENO schemes. *J. Comput. Phys.* **228**, 8481–8524 (2009)
15. Glimm, J., Grove, J., Li, X., Oh, W., Tan, D.C.: The dynamics of bubble growth for Rayleigh–Taylor unstable interfaces. *Phys. Fluids* **31**, 447–465 (1988)
16. Gu, Y., Gao, Z., Hu, G., Li, P., Wang, L.: High order finite difference alternative WENO scheme for multi-component flows. *J. Sci. Comput.* **89**, 52 (2021)
17. Kim, C.H., Ha, Y., Yoon, J.: Modified nonlinear weights for fifth-order weighted essentially non-oscillatory schemes. *J. Sci. Comput.* **299–323**, 67 (2016)
18. Ha, Y., Kim, C.H., Yang, H., Yoon, J.: Sixth-order weighted essentially non-oscillatory schemes based on exponential polynomials. *SIAM J. Sci. Comput.* **38**(4), A1987–A2017 (2016)
19. Ha, Y., Kim, C.H., Yang, H., Yoon, J.: Improving accuracy of the fifth-order WENO scheme by using the exponential approximation space. *SIAM J. Numer. Anal.* **59**, 143–172 (2021)
20. Ha, Y., Kim, C.H., Lee, Y.J., Yoon, J.: An improved weighted essentially non-oscillatory scheme with a new smoothness indicator. *J. Comput. Phys.* **232**, 68–86 (2013)
21. Harten, A., Osher, S.: Uniformly high-order accurate non-oscillatory schemes I. *SIAM J. Numer. Anal.* **24**, 279–309 (1987)
22. Harten, A., Engquist, B., Osher, S., Chakravarthy, S.: Uniformly high-order accurate non-oscillatory schemes III. *J. Comput. Phys.* **71**, 231–303 (1987)
23. Hesthaven, J.S., Mönkeberg, F.: Entropy stable essentially nonoscillatory methods based on RBF reconstruction. *ESAIM: M2AN* **53**, 925–958 (2019)
24. Hesthaven, J.S., Mönkeberg, F.: Two-dimensional RBF-ENO method on unstructured grids. *J. Sci. Comput.* **82**, 1–24 (2020)
25. Hesthaven, J.S., Mönkeberg, F.: Hybrid high-resolution RBF-ENO method. *J. Comput. Phys. X* **12**, 100089 (2021)
26. Henrick, A.K., Aslam, T.D., Powers, J.M.: Mapped weighted-essentially-non-oscillatory schemes: achieving optimal order near critical points. *J. Comput. Phys.* **207**, 542–567 (2005)
27. Jeong, B., Yang, H., Yoon, J.: Development of a WENO scheme based on radial basis function with an improved convergence order. *J. Comput. Phys.* **468**, 111502 (2022)
28. Jiang, G., Shu, C.-W.: Efficient implementation of weighted ENO schemes. *J. Comput. Phys.* **126**, 202–228 (1996)
29. Jiang, Y., Shu, C.-W., Zhang, M.: An alternative formulation of finite difference WENO schemes with Lax–Wendroff time discretization for conservation laws. *SIAM J. Sci. Comput.* **35**, A1137–A1160 (2013)
30. Jiang, Y., Shu, C.-W., Zhang, M.: Free-stream preserving finite difference schemes on curvilinear meshes. *Methods Appl. Anal.* **21**, 1–30 (2014)
31. Hu, X.Y., Wang, Q., Adams, N.A.: An adaptive central-upwind weighted essentially non-oscillatory scheme. *J. Comput. Phys.* **229**, 8952–8965 (2010)
32. Hu, X.Y., Adams, N.A.: Scale separation for implicit large eddy simulation. *J. Comput. Phys.* **230**, 7240–7249 (2011)
33. Käser, M., Iske, A.: ADER schemes on adaptive triangular meshes for scalar conservation laws. *J. Comput. Phys.* **205**, 486–508 (2005)
34. Lax, P.D.: Weak solutions of nonlinear hyperbolic equations and their numerical computation. *Commun. Pure Appl. Math.* **7**, 159–193 (1954)
35. Levy, D., Puppo, G., Russo, G.: Central WENO schemes for hyperbolic systems of conservation laws. *M2AN Math. Model. Numer. Anal.* **33**, 547–571 (1999)
36. Levy, D., Puppo, G., Russo, G.: Compact central WENO schemes for multidimensional conservation laws. *SIAM J. Sci. Comput.* **22**, 656–672 (2000)
37. Li, P., Li, T., Don, W.-S., Wang, B.-S.: Scale-invariant multi-resolution alternative WENO scheme for the Euler equations. *J. Sci. Comput.* **94**, 15 (2023)
38. Liu, H.: A numerical study of the performance of alternative weighted ENO methods based on various numerical fluxes for conservation law. *Appl. Math. Comput.* **296**, 182–197 (2017)
39. Li, G., Qiu, J.: Hybrid weighted essentially non-oscillatory schemes with different indicators. *J. Comput. Phys.* **229**, 8105–8129 (2010)

40. Liu, H., Qiu, J.: Finite difference Hermite WENO schemes for conservation laws, II: an alternative approach. *J. Sci. Comput.* **66**, 598–624 (2016)
41. Liu, X.-D., Osher, S., Chan, T.: Weighted essentially non-oscillatory schemes. *J. Comput. Phys.* **115**, 200–212 (1994)
42. Liu, H.X., Jiao, X.M.: WLS-ENO: weighted-least-squares based essentially non-oscillatory schemes for finite volume methods on unstructured meshes. *J. Comput. Phys.* **314**, 749–773 (2016)
43. Pirozzoli, S.: Conservative hybrid compact-WENO schemes for shock-turbulence interaction. *J. Comput. Phys.* **178**, 81–117 (2002)
44. Schulz-Rinne, C.W., Collins, J.P., Glaz, H.M.: Numerical solution of the Riemann problem for two-dimensional gas dynamics. *SIAM J. Sci. Comput.* **14**, 1394–1414 (1993)
45. Shen, Y.Q., Yang, G.W.: Hybrid finite compact-WENO schemes for shock calculation. *Int. J. Numer. Methods Fluids* **53**, 531–560 (2007)
46. Shi, J., Zhang, Y.-T., Shu, C.-W.: Resolution of high order WENO schemes for complicated flow structures. *J. Comput. Phys.* **186**, 690–696 (2003)
47. Shi, Y., Guo, Y.: A fifth order alternative compact-WENO finite difference scheme for compressible Euler equations. *J. Comput. Phys.* **397**, 108873 (2019)
48. Shu, C.-W., Osher, S.: Efficient implementation of essentially non-oscillatory shock capturing schemes. *J. Comput. Phys.* **77**, 439–471 (1988)
49. Shu, C.-W., Osher, S.: Efficient implementation of essentially non-oscillatory shock capturing schemes II. *J. Comput. Phys.* **83**, 32–78 (1989)
50. Sod, G.: A survey of several finite difference methods for systems of nonlinear hyperbolic conservation laws. *J. Comput. Phys.* **27**, 1–31 (1978)
51. Titarev, V.A., Toro, E.F.: Finite volume WENO schemes for three-dimensional conservation laws. *J. Comput. Phys.* **201**, 238–260 (2014)
52. Toro, E.F.: *Riemann solvers and numerical methods for fluid dynamics*. Springer, New York (1997)
53. Toro, E.F., Spruce, M., Speares, W.: Restoration of the contact surface in the HLL-Riemann solver. *Shock Waves* **4**, 25–34 (1994)
54. Wang, B.-S., Don, W.S., Garg, N.K., Kurganov, A.: Fifth-order A-WENO finite-difference schemes based on a new adaptive diffusion central numerical flux. *SIAM J. Sci. Comput.* **42**, A3932–A3956 (2020)
55. Wang, B.-S., Li, P., Gao, Z., Don, W.S.: An improved fifth order alternative WENO-Z finite difference scheme for hyperbolic conservation laws. *J. Comput. Phys.* **374**, 469–477 (2018)
56. Woodward, P., Colella, P.: The numerical simulation of two-dimensional fluid flow with strong shocks. *J. Comput. Phys.* **54**, 115–173 (1984)
57. Xu, Z.F., Shu, C.-W.: Anti-diffusive flux corrections for high order finite difference WENO schemes. *J. Comput. Phys.* **205**, 458–485 (2005)
58. Young, Y.-N., Tufo, H., Dubey, A., Rosner, R.: On the miscible Rayleigh–Taylor instability: two and three dimensions. *J. Fluid Mech.* **447**, 377–408 (2001)
59. Zhang, R., Zhang, M., Shu, C.-W.: On the order of accuracy and numerical performance of two classes of finite volume WENO schemes. *Commun. Comput. Phys.* **5**, 836–848 (2009)
60. Zhu, J., Qiu, J.: Trigonometric WENO schemes for hyperbolic conservation laws and highly oscillatory problems. *Commun. Comput. Phys.* **8**, 1242–1263 (2010)
61. Zhu, J., Qiu, J.: WENO schemes and their application as limiters for RKDG methods based on trigonometric approximation spaces. *J. Sci. Comput.* **55**, 606–644 (2013)
62. Zhu, J., Qiu, J.X.: A new fifth order finite difference WENO scheme for solving hyperbolic conservation laws. *J. Comput. Phys.* **318**, 110–121 (2016)
63. Zhu, J., Qiu, J.X.: A new type of finite volume WENO schemes for hyperbolic conservation laws. *J. Sci. Comput.* **73**, 1338–1359 (2017)

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